

GENERAL THETA FOUNDATIONS I: COMPATIBLE STATISTICAL EXPERIMENTS AND RESOURCE FRONTIERS

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ABSTRACT. We study statistical experiments subject to a register constraint at each cut of a finite causal protocol. Their feasible occupation arrays are characterized by linear flow equations and quadratic compatibility equations, including at zero-probability histories. This yields constructive realizations and a complete polynomial upper-certificate hierarchy for the full-profile minimax value, without convexifying the class of machines. For binary experiments we obtain an exact posterior partition recursion for Bayesian decisions and an exact receiver-curve recursion for minimax decisions with binary registers. In a three-report experiment with overlapping binary noise, we compute the entire register-profile frontier for both objectives; the resulting strict gap shows that a free mixture of machines changes the minimax value. The characterization is connected to finite-precision response-cover auditors, acquired calibration, and positive-noise physical experiments, with preparations, intermediate registers and clocks explicitly charged. The marked continuation, residual, microscopic and Gaussian comparison results are developed with their proofs and with their distinct hypotheses retained.

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1. INTRODUCTION

An experiment is used through a causal observer. Information acquired at one time may be lost at a later checkpoint, and a decision rule that is economical after training may be expensive to train. The statistical value of an observer therefore depends on its entire register profile, not only on the dimension of its last statistic. A further difficulty is that two feasible observers need not be mixable at the same profile: the variable that selects an observer has to be retained wherever their future actions differ.

We study this problem from positive experimental kernels, preparations, causal interventions and bounded executable tests. These objects do not presuppose a convex parameter image, an information metric, or a finite-dimensional closure of the underlying dynamics. The present paper develops their finite resource layer and its comparison maps. The word “finite” refers to a declared protocol and its observable alphabets, not to a removal of the microscopic or limiting questions in the larger theory.

1.1. The full-profile problem. Fix a finite schedule σ , a register profile $\mathbf{K}$, a family of environments B , and a terminal payoff $G \in [-1, 1]$. Let

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G) = \sup_A \inf_{b \in B} \mathbb{E}_b^A G,$$

where every persistent data-dependent variable of A passes through the specified registers. The clocked model allows a predetermined transition table at each epoch; its clock carries no discarded data. An autonomous controller instead reuses its declared transition rows and must retain any phase variable needed to select a different row. Refining an atomic operation into input-bit or random-bit epochs changes σ and its resource problem. These conventions are part of the theorem statements.

The central object is a compatible occupation array. Its coordinates record the probability mass entering and leaving each observable controller row, under every environment and hidden history. Ordinary flow conservation is necessary but not sufficient. All hidden histories that present the same observable row must use the same stochastic transition. This requirement is a collection of quadratic proportionality equations. Unlike a division-based parametrization, these equations also retain every zero-occupation boundary case.

Theorem 1.1 (Compatible statistical resource frontiers). *The following statements hold for the indicated finite protocols.*

- (i) *For a finite environment family, compatible occupation arrays are exactly the arrays of all stochastic machines of the full profile. Their minimax value is attained. It equals the decreasing limit of polynomial sum-of-squares upper certificates, and every feasible array reconstructs a machine. Shared-row equations give the corresponding autonomous characterization. Finite-precision controller witnesses and compact-family kernel nets give matching two-sided approximations.*
- (ii) *For binary, conditionally independent finite observations, possibly with a finite controlled choice of experiment, the optimal clocked Bayesian value at any finite register profile is given by a recursion of compatible interval partitions of incoming posterior atoms. For uncontrolled observations with one- or two-state registers, a receiver-curve recursion gives the exact clocked minimax value at every finite horizon. These are optimizations over all stochastic machines, not implementation bounds for a chosen statistic.*
- (iii) *With three independent binary reports of error probability $p \in (0, 1/2)$, put $d = p(1-p)(1-2p)$. At the atomic profile $(2, 2, 2)$ the exact fair-prior and minimax successes are respectively $1 - p + d/2$ and $1 - p + d/3$. At profile $(2, 3, 2)$ both equal $1 - p + d$. A free mixture of the two Bayesian optimal machines would raise the first profile's minimax value by $d/6$, so that convexification is strictly incorrect for that resource problem.*
- (iv) *A fixed K -response calibration can be acquired from $M = 2^s$ target preparations, with uniform error at most ρ except with probability β , whenever $M \geq (2\rho^2)^{-1} \log(2K/\beta)$. Its $(M + 1)^K$ -state counter vector is retained and priced through the subsequent auditor. The resulting response-cover constructions and their physical consumers belong to the same full-profile classes as the occupation converse.*

Proof. Part (i) is Theorems 2.1 and 2.2 and Corollary 2.3. Part (ii) is Theorems 3.2 and 3.3. Part (iii) is Theorem 4.1 and Corollary 4.3. Part (iv) is Theorem 5.2 and Corollary 6.1. $\square$

The certificate hierarchy in (i) uses a classical positivity theorem. Its role is to give an upper bound on every complete machine, paired with the reverse reconstruction from feasible occupations. It is not an efficiency or finite-termination assertion. The noisy recursions in (ii) and the closed formulas in (iii) supply structural information that a generic polynomial encoding alone would not provide. In particular, the distinction between Bayesian and minimax objectives survives in an experiment whose two laws have common support at every observation.

1.2. Comparison with existing methods. Comparison of experiments is classical [1, 2, 13]. Filtered randomization is also an existing subject. Weisshaupt's filtered criterion [10, Lemma 1 and Theorem 5] concerns compact convex classes of positive operators and a risk criterion for filtered randomization. Our feasible object is a specified finite register execution, whose behavior set need not be convex. We also cite Norberg [3]; the original proof text has not been obtained for a definitive theorem-level comparison, and no priority claim rests on that gap in access.

Finite-memory statistical learning and the role of time-dependent transition rules are treated by Cover and by Hellman and Cover [24, 26]. Likelihood-ratio quantizer optimality is a classical mechanism [27, Section III]. We use it both for compatible posterior partitions and for binary-channel domination. The receiver-curve and equal-error arguments are not introduced as new

statistical principles. Their role here is to solve one full-profile optimization, including its stochastic minimax part and the selector cost exposed by the explicit three-report frontier.

Polynomial descriptions of partially observed controller occupations are likewise established. Müller and Montúfar [28, Example 15, Theorem 16 and Remark 19] give semialgebraic feasible-frequency descriptions and polynomial planning for infinite-horizon memoryless POMDPs, with their stated positivity assumptions. The finite-protocol formulation here makes the observable row equivalences, changing register alphabets and zero-mass reconstruction explicit. Its upper-certificate completeness applies Putinar’s theorem and its rational version [29, 30]. The combined contribution is the exact compatible frontier, its noisy evaluations, and its connections to fully priced experimental consumers; neither polynomial optimization nor the vanishing-minor identity is claimed as a new invention.

1.3. Organization and scope of the applications. Sections 2–6 contain the principal proof chain. The marked continuation and decision spectra then identify the underlying experimental interfaces. The finite precision, response-cover tournaments, multicut revelation formula, weighted residuals, nonlinear tests and adaptive tests are developed with their proofs. The microscopic moving-collision construction and the bounded Gaussian comparison produce physical consumers of the same interfaces. The original short, polynomial-horizon and long-clock physical resource points are all retained.

Those three points are constructions, not an exact physical Pareto curve. The general converse now applies to each of their fixed protocols, but its numerical evaluation across all physical budgets is a separate task. Likewise a confidence lower bound at a specified retained-coin interface is not automatically a lower bound for an auditor that uses richer raw observations. These distinctions are mathematical typing conditions on the applications, not changes to the physical or operator targets.

The article retains the full development of its supporting results; versioned derivation and review records are kept outside the exposition. The dependency appendix records the namespaced eleven-component program. The primary A2 geometric chain remains independent, while B4 kinetic and broader C2 operator conclusions require their own stated hypotheses and proofs. No artificial dependence on a finite-state theorem is imposed on those branches of the program.

2. COMPATIBLE OCCUPATION ARRAYS

A fixed terminal statistic and an optimal statistic under a sequence of memory constraints are different objects. We begin with a characterization of the second object. All the spaces in this section are finite. No strict positivity assumption is imposed on a probability coordinate.

2.1. The finite protocol and its observable rows. Fix a finite schedule σ and a register profile $\mathbf{K} = (K_t)_{t \in \sigma}$. A cut is part of the specification of σ : subdividing an atomic observation or a randomization operation inserts additional register constraints. The clocked model permits a different transition table at each scheduled operation. It does not permit access to a discarded observation. We write $\mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})$ for this class. Initial randomization, when allowed, is a transition of the same kind as all other randomizations.

Let Θ be a finite set of environments. Under θ , all preparation and experimental kernels are specified. One may include the complete physical and public past in a finite hidden index i . This index is a variable in the analysis, not an observer register. An elementary observer operation has an accessible row r and chooses j from a nonempty finite set J_r . The row contains exactly the current register and the current atomic input, if there is one. The choice j can specify a new register and a public action. A public action that is subsequently needed by the observer must be

retained in that register. Every permissible transition has the form

$$u(j \mid r), \quad u(j \mid r) \geq 0, \quad \sum_{j \in J_r} u(j \mid r) = 1. \quad (2.1)$$

Neither θ nor the unobserved part of i is an argument of u .

Known environment operations act linearly on conditional occupation masses. For example, if $z(\theta, h, m)$ is the mass before an observation, then its incoming mass at the observer is

$$x(\theta, h', (m, y)) = \sum_h z(\theta, h, m) L^\theta(h', y \mid h, m). \quad (2.2)$$

Dependence on a previously issued action is included in h or in the retained part of m . After an observer choice, the public history is updated and discarded indices are summed out, again by a known linear map. Alternating these two kinds of operation represents a finite controlled experiment, including feedback, calibration, validation, and a terminal readout. The cut alphabets and the allowed rows are fixed before optimization. In particular, this representation does not silently grant the observer the hidden history used to define the experimental kernels.

For each observer operation introduce outgoing masses $f(\theta, i, r, j)$. Impose the conservation and compatibility equations

$$f(\theta, i, r, j) \geq 0, \quad \sum_{j \in J_r} f(\theta, i, r, j) = x(\theta, i, r), \quad (2.3)$$

$$f(\theta, i, r, j) x(\theta', i', r) = f(\theta', i', r, j) x(\theta, i, r) \quad (\theta, i, \theta', i', r, j). \quad (2.4)$$

All incoming masses are nonnegative and are linked to the preceding operation by the known linear maps. Preparations are prescribed and have conditional mass one for each θ . We call the resulting set of arrays $\mathcal{O}_\sigma^{\text{clk}}(\mathbf{K})$.

Theorem 2.1 (Exact full-profile realization). *The occupation arrays of $\mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})$ are exactly $\mathcal{O}_\sigma^{\text{clk}}(\mathbf{K})$. This set is compact and is defined by linear equations, nonnegativity, and the quadratic equations (2.4). Every feasible array determines an implementing machine, with arbitrary legal rows only where the incoming mass vanishes under every environment.*

More generally, suppose specified occurrences of observable rows must use the same transition distribution, with their choices identified. For each such class impose (2.4) also between different occurrences. The resulting arrays are exactly those realized by machines with these shared rows. In particular this gives an exact formulation for an autonomous controller with a fixed state alphabet and a single reused transition table.

Proof. A machine gives $f(\theta, i, r, j) = x(\theta, i, r) u(j \mid r)$, so conservation and every minor equation follow. Conversely, fix an occurrence of r . If some $x(\theta_0, i_0, r) > 0$, define

$$u(j \mid r) = \frac{f(\theta_0, i_0, r, j)}{x(\theta_0, i_0, r)}.$$

Conservation makes this a stochastic row. The minor with (θ_0, i_0) shows that the same formula reproduces f at every positive incoming mass. At a zero incoming mass, nonnegativity and conservation force each outgoing mass to be zero. If all incoming masses vanish, choose any legal stochastic row. Thus the recovered row is independent of both the environment and the hidden index.

Perform this construction at every observer operation. Induction along the schedule, using (2.2) and the corresponding linear outgoing maps, proves that the recovered machine has exactly the given occupation arrays. Its registers are the declared alphabets; the proof has not introduced a retained selector. For a shared row class, choose one positive incoming mass anywhere in that class and use the cross-occurrence equations. The same argument reconstructs one common row. A class with no positive incoming mass is treated arbitrarily.

Each coordinate is a conditional probability or a subprobability mass and is therefore at most one. Adding these redundant upper bounds makes boundedness explicit. The equations and inequalities are closed, and there are finitely many coordinates. Compactness follows. $\square$

The zero-mass provision is essential. Cancelling an occupation coordinate without a case distinction would remove valid boundary machines. Nor would separate low-rank descriptions at successive cuts imply (2.2)–(2.4). The latter equations refer to one execution, not to a list of independently selected encoders. For an autonomous machine, the phase cannot be supplied through a row name unless it is either an actual input or part of the charged state. A clocked machine can be embedded autonomously by retaining (t, m) ; Theorem 2.1 also permits direct optimization without making that possibly wasteful construction.

2.2. A complete polynomial converse. Let $G \in [-1, 1]$ be a terminal payoff and let $L_\theta(z)$ denote its expectation as a linear function of the complete occupation array z . The terminal decision is included among the observer operations. Set

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(\Theta, G) = \max_{A \in \mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})} \min_{\theta \in \Theta} \mathbb{E}_\theta^A G. \quad (2.5)$$

The admissible row sets are assumed nonempty. Introduce $v \in [-1, 1]$ and impose $L_\theta(z) - v \geq 0$ for every θ . Write $h_a = 0$ for all the linear and quadratic equalities and $g_b \geq 0$ for the remaining inequalities. Include the redundant ball inequality

$$R - \|(z, v)\|^2 \geq 0,$$

where the positive integer R is larger than the number of coordinates. Let $\mathcal{F}$ be this basic closed feasible set. For $d \geq 1$, define U_d as the infimum of $u \in \mathbb{R}$ for which the polynomial identity

$$u - v = \sigma_0 + \sum_b \sigma_b g_b + \sum_a q_a h_a \quad (2.6)$$

holds, with each σ_b a sum of squares, each q_a a polynomial, and every displayed summand of degree at most $2d$. An empty certificate class has infimum $+\infty$.

Theorem 2.2 (Complete full-profile certificates). *For the finite protocol above,*

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(\Theta, G) = \max_{(z, v) \in \mathcal{F}} v = \inf_{d \geq 1} U_d, \quad U_{d+1} \leq U_d. \quad (2.7)$$

A feasible occupation array is a constructive lower certificate. An identity (2.6) is an upper certificate for every stochastic machine of the specified full profile. For rational experimental data, every rational $u > V_{\sigma, \mathbf{K}}^{\text{clk}}$ admits such a certificate with rational coefficients. The same conclusions hold with shared rows, including the autonomous case.

Let ℓ_r be the best worst-case payoff among the actual controller tables whose entries are dyadic with denominator 2^r . If transition t has at most s_t choices per row, then

$$\ell_r \leq V_{\sigma, \mathbf{K}}^{\text{clk}} \leq \min \left\{ U_d, \ell_r + 2 \sum_t (s_t - 1) 2^{-r} \right\}. \quad (2.8)$$

Here repeated uses of a shared row are counted repeatedly in the sum. Thus there are matching constructive and converse certificates to any positive accuracy, without a minimax interchange or a convexification of fixed-width machines.

Proof. The first equality follows from Theorem 2.1 and the inequalities $v \leq L_\theta(z)$. Compactness gives attainment. Evaluating (2.6) at any feasible point gives $v \leq u$. Increasing d only enlarges the certificate class.

For completeness, take $u > \max_{\mathcal{F}} v$. The polynomial $u - v$ is strictly positive on $\mathcal{F}$. Apply Putinar's positivity theorem [29] to the inequalities $g_b \geq 0$ and the pairs $h_a \geq 0, -h_a \geq 0$. The explicitly included ball makes the quadratic module Archimedean. The resulting identity has

finite degree. Collecting the two multipliers of h_a gives (2.6). Letting u decrease to the maximum proves the second equality. For rational data and rational u , the rational version with an explicit ball generator [30, Theorem 6] gives rational sums of squares and rational multipliers. This is an application of those positivity theorems, not a new Positivstellensatz.

Finally round each stochastic row to the dyadic simplex, preserving its causal zeros, at total-variation cost at most $(s_t - 1)2^{-r}$. Round a shared row once and use that rounded row at every occurrence. Coupling the original and rounded executions until their first discrepancy bounds the probability of a discrepancy by the sum in (2.8), capped at one. The payoff range is at most two. This estimate is uniform in θ , so rounding a maximizing machine proves the rightmost bound. The dyadic machines are actual machines, which proves the left bound. Their finite enumeration is a construction, even though its cost may be large. $\square$

A numerical semidefinite solution is not by itself an identity (2.6); coefficients and nonnegativity must be certified. The theorem proves convergence, not a uniform degree bound, finite termination at $u = V$, or an efficient algorithm. Its necessity statement is nevertheless different from a resource upper construction: every machine obeys every certified upper bound. Dropping compatibility constraints produces a relaxation. It does not preserve the exact value in general, as Theorem 4.1 will demonstrate explicitly.

Corollary 2.3 (Compact families and finite experimental adapters). *Suppose B is a compact family of finite experimental kernels on a fixed schedule with N environment calls. Let $B_j \subset B$ be a finite ω_j -net in the maximum conditional total-variation metric, with $\omega_j \rightarrow 0$. Use the same bounded terminal payoff throughout. Then*

$$\ell_r(B_j) - 2N\omega_j \leq V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G) \leq U_d(B_j). \quad (2.9)$$

As j, r, d tend to infinity in a suitable order, the two sides have common limit $V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G)$. A certified adapter changing call i by at most δ_i contributes $2\sum_i \delta_i$ to a payoff comparison. A simulator with profile $\mathbf{R}$ is charged by the product profile $(R_t K_t)$.

Proof. For any one controller, couple corresponding calls until the first unequal outcome. Its expected payoff varies by at most $2N\omega_j$ between a kernel and its net representative. Hence $V(B_j) - 2N\omega_j \leq V(B) \leq V(B_j)$. Apply Theorem 2.2 to B_j and choose the dyadic and certificate accuracies after choosing j . The same coupling with unequal δ_i gives the adapter bound. Retaining the ordered pair of simulator and controller states proves the product assertion. $\square$

An effective net is additional data; abstract compactness is not an algorithm for an arbitrary presentation. The finite-alphabet, compact parameter collision experiments considered below have this interface. Consequently their fixed-schedule, full-profile values admit the same all-auditor certificates. This implication does not evaluate those certificates numerically for every physical sample budget, nor does it identify a minimal autonomous implementation.

3. CONTINUATION POLYTOPES AND NOISY POSTERIOR PARTITIONS

We next obtain a finite exact recursion, and then an explicit solution, for noisy experiments. Unlike a revelation channel, these experiments need never disclose the parameter at any time.

3.1. One common continuation matrix. Let Θ and the terminal action set A be finite. Conditional on θ , report t has distribution p_t^θ on Y_t and reports are independent. A clocked observer starts in one state, retains at most K_t states after report t , and receives payoff $G(\theta, a)$. For the moment there are no prescribed zeros in its update tables. For a fixed prior π , denote its optimal Bayesian payoff by $B_{n, \mathbf{K}}^{\text{clk}}(\pi, G)$. The terminal readout has no further observation of a discarded report.

Define finite sets of continuation matrices, with one column per retained state, backwards by

$$\mathcal{C}_n = \{(G(\theta, a_s))_{\theta, s} : (a_s)_s \in A^{K_n}\}, \quad (3.1)$$

$$\mathcal{C}_{t-1} = \bigcup_{f: [K_{t-1}] \times Y_t \rightarrow [K_t]} T_{t,f}(\mathcal{C}_t), \quad (3.2)$$

$$(T_{t,f}C)_{\theta, s} = \sum_{y \in Y_t} p_t^\theta(y) C_{\theta, f(s, y)}, \quad K_0 = 1. \quad (3.3)$$

The same matrix C occurs in every term of (3.3). Different incoming states do not receive independently selected future machines. If there are finitely many controlled experiments, include an action choice a_s in the union and replace $p_t^\theta(y)$ by $p_t^\theta(y \mid a_s)$.

Proposition 3.1 (Exact Bayesian continuation recursion). *Every matrix in $\mathcal{C}_t$ is implemented by one deterministic suffix of the prescribed profile, and every such suffix has a matrix in $\mathcal{C}_t$. Moreover,*

$$B_{n, \mathbf{K}}^{\text{clk}}(\pi, G) = \max_{C \in \mathcal{C}_0} \sum_{\theta} \pi_{\theta} C_{\theta, 1}. \quad (3.4)$$

For Bayesian optimization the sets can be replaced by their joint convex hulls $\mathcal{P}_t = \text{conv } \mathcal{C}_t$, using $\mathcal{P}_{t-1} = \text{conv } \bigcup_f T_{t,f}(\mathcal{P}_t)$. This operation does not authorize a free shared random seed. Replacing $\mathcal{P}_t$ by the product of its separate column projections is not valid in general.

Proof. At the terminal time, each column records the selected terminal action. At an earlier time, a deterministic update f feeds the new state into one fixed suffix. Conditional expectation gives (3.3). Backward induction proves the realization and converse statements, including recovery of the update tables.

For a fixed prior the payoff is a multiaffine function of the finitely many time-indexed stochastic rows: an execution visits any such row at most once. A maximum exists. Holding all but one row fixed, replace that row by a maximizing vertex of its simplex. Doing this finitely many times gives a deterministic maximum without changing the budget. The same argument includes an experiment-selection row. This proves (3.4) for all stochastic machines. Linearity of $T_{t,f}$ gives the asserted convex-hull recursion. A maximizing vertex of the final finite hull has a deterministic witness; the hull is used only to calculate a support value.

For the last assertion, consider a suffix with two possible initial states, a compulsory next cut of width one, and no informative subsequent report. Its two continuation columns must be equal. Individually each column may implement either constant binary decision. The product of the two column projections also contains the matrix that makes opposite decisions in the two initial states, which the compulsory one-state cut cannot implement. Under a fair preparation where the two initial states reveal the binary parameter, this false matrix gives success one instead of the true value one half. $\square$

3.2. Binary posterior geometry. For $\Theta = \{0, 1\}$ write a current occupation list as $Z = (z_s)_s$, where

$$z_s = (\mathbb{P}(\theta = 0, M = s), \mathbb{P}(\theta = 1, M = s)) \in \mathbb{R}_+^2.$$

Zero columns can be discarded and unused labels can always be padded. After report t , its incoming atoms are

$$b_{s,y} = (z_{s,0} p_t^0(y), z_{s,1} p_t^1(y)). \quad (3.5)$$

Order the nonzero atoms by $q(b) = b_1/(b_0 + b_1) \in [0, 1]$. An interval partition divides that ordered list into at most K_t consecutive nonempty blocks; summing the atoms in each block gives a new list $\mathcal{I}_t(Z)$. Equal-posterior atoms may first be merged. No nestedness of the partitions at different times is required.

Set

$$\begin{aligned} F_n(Z) &= \sum_s \max_{a \in A} \{z_{s,0}G(0, a) + z_{s,1}G(1, a)\}, \\ F_{t-1}(Z) &= \max_{\mathcal{I}_t} F_t(\mathcal{I}_t(Z)), \quad t = n, \dots, 1. \end{aligned} \tag{3.6}$$

For controlled observations, first maximize over the finite vector of experiment choices $(a_s)_s$ and then form (3.5) with those choices. Only lists reachable from the initial list are needed in evaluating this recursion.

Theorem 3.2 (Exact noisy full-profile recursion). *For arbitrary binary finite experiments of the conditionally independent form above, any prior, any bounded terminal payoff, and any finite cardinality profile,*

$$B_{n, \mathbf{K}}^{\text{clk}}(\pi, G) = F_0((\pi_0, \pi_1)). \tag{3.7}$$

There is an optimal deterministic machine whose update at every time is an interval partition of the current incoming posterior atoms. The recursion produces such a machine and optimizes over all stochastic machines, not just over a preselected statistic. It also holds with a finite choice of controlled experiments at each state.

Proof. Deterministic optimality follows from Proposition 3.1. Fix a deterministic global maximum, and consider one of its update times. Hold its prefix, experiment choices at this time, and entire suffix fixed. If the suffix starting at label j has continuation rewards c_{0j}, c_{1j} , assigning incoming atom b to j contributes

$$b_0 c_{0j} + b_1 c_{1j} = (b_0 + b_1)((1 - q(b))c_{0j} + q(b)c_{1j}).$$

It is therefore optimal to select a maximizing affine function of q . The upper envelope of finitely many affine functions on $[0, 1]$ has interval regions. Choose tied affine functions consistently, ordering active slopes and discarding identical duplicates. Each selected label then occupies an interval. At an identical posterior, all atoms have the same maximizing set, so they can be assigned to the same label. Unused labels cause no difficulty. Relabel the active suffix states in interval order, including their future experiment choices.

Now perform these replacements in chronological order. A replacement does not alter any earlier incoming occupation list or earlier update. It cannot decrease the payoff and thus preserves global optimality. Later incoming lists may change; they are recomputed before their own replacement. Hence the final machine has the interval property at all times simultaneously. This forward argument is why independently optimizing a partition at each cut would not prove the theorem.

Conversely, every sequence of choices in (3.6) defines causal finite update tables and therefore one admissible machine. The terminal formula is exactly its optimal readout. The existence just proved shows that restricting the finite search to these choices loses nothing. This proves (3.7). With controlled observations, first keep the current deterministic experiment choices fixed in the replacement argument; the same affine-envelope reasoning applies to their incoming atoms. Maximizing over those finite choices completes the proof. $\square$

This theorem uses the likelihood-ratio quantizer principle [27]; the affine-envelope mechanism is not claimed as new. The point of the recursion is the common, changing occupation list and the simultaneous realizability of all its partitions. The posterior pairs are computed by the designer. The implemented machine stores only a finite label and uses a fixed finite lookup table; it does not store a real-valued posterior. The number of reachable lists can be large. No polynomial-time claim is part of the theorem.

The result is Bayesian and clocked. A maximum of a minimum over environments is not multiaffine, and a reused autonomous row can occur several times in one execution. Neither of those cases is covered by the purification argument. They remain covered by Theorems 2.1 and

2.2. The following example solves the minimax problem directly and shows why the distinction is substantive.

3.3. An exact minimax recursion for binary registers. There is also a structural recursion that does not purify the minimax objective. In this subsection observations are conditionally independent and uncontrolled, and every retained alphabet has size one or two. Represent a binary prefix experiment by its pair $r = (r_0, r_1)$, where $r_\theta = \mathbb{P}_\theta(M = 1)$. Its labels may be exchanged at no cost. Say that s dominates r if a single stochastic binary channel, independent of θ , maps s to r .

For a pair s and the next report law, let $E_t(s)$ be the experiment on $\{0, 1\} \times Y_t$ with masses

$$e_\theta(m, y) = s_\theta^m (1 - s_\theta)^{1-m} p_t^\theta(y).$$

At endpoints this formula has its direct Bernoulli meaning. For any finite binary experiment $E = (e_0, e_1)$ define its upper receiver curve by the finite linear program

$$R_E(a) = \max \left\{ \sum_x e_1(x) \phi(x) : 0 \leq \phi(x) \leq 1, \sum_x e_0(x) \phi(x) = a \right\}, \quad 0 \leq a \leq 1. \quad (3.8)$$

Ordering atoms by likelihood ratio solves this program: accept the largest ratios first, randomizing at at most one ratio level. Atoms with zero e_0 are treated by their direct contribution to the objective; no division by zero is needed.

Starting with $\mathcal{R}_0 = \{(0, 0)\}$, define

$$\mathcal{R}_t = \begin{cases} \{(0, 0)\}, & K_t = 1, \\ \{(a, R_{E_t(s)}(a)) : s \in \mathcal{R}_{t-1}, 0 \leq a \leq 1\}, & K_t = 2. \end{cases} \quad (3.9)$$

This is a recursion of attainable channels, not of their convex hulls. Its continuous parameters are design parameters of finite stochastic rows, not stored posterior coordinates. In particular, it need not have a polygonal reachable set after several stages.

Theorem 3.3 (Binary-register minimax frontier). *Every channel in $\mathcal{R}_t$ is attainable with the prefix profile, and every stochastic prefix experiment with that profile is dominated by some member of $\mathcal{R}_t$. For a bounded payoff G and finite terminal action set, let $\Gamma_G(r)$ be the optimal minimum of the two expected payoffs obtained by a terminal stochastic readout of the binary experiment r . Then*

$$V_{n, \mathbf{K}}^{\text{clk}}(\{0, 1\}, G) = \max_{r \in \mathcal{R}_n} \Gamma_G(r). \quad (3.10)$$

For correct-decision payoff this reduces to

$$V_{n, \mathbf{K}}^{\text{clk}} = \max_{r \in \mathcal{R}_n} \left\{ \frac{1}{2} + \frac{|r_1 - r_0|}{2(1 + |1 - r_0 - r_1|)} \right\}. \quad (3.11)$$

An optimizer can be implemented by likelihood-ratio quantizers with at most one randomized ratio level per update and a terminal readout. These assertions hold at every finite horizon, with overlapping noise and with arbitrary placement of one-state cuts.

Proof. First consider a binary summary of any finite binary experiment, with output probabilities (r_0, r_1) . Exchange its labels if necessary so that $r_1 \geq r_0$. The receiver-curve optimizer at $a = r_0$ has output probabilities (r_0, b) with $b \geq r_1$. It dominates (r_0, r_1) . Indeed, if $b > r_0$, put

$$\lambda = \frac{r_1 - r_0}{b - r_0} \in [0, 1].$$

The postprocessing probabilities $\mathbb{P}(M = 1 \mid S = 0) = r_0(1 - \lambda)$ and $\mathbb{P}(M = 1 \mid S = 1) = r_0(1 - \lambda) + \lambda$ are in $[0, 1]$ and give the required pair. If $b = r_0$, the desired summary is uninformative and can be generated by a constant stochastic row. Undoing the exchange of labels covers the original summary. The receiver program is solved by an ordered threshold because moving an equal amount

of e_0 -mass from a smaller likelihood ratio to a larger one cannot decrease its objective. Sorting the finite atoms and filling the prescribed mass a proves the stated one-level randomization rule, including zero-mass atoms.

Proceed by induction on t . Each member of (3.9) is attained by an attaining previous channel and a receiver-curve quantizer. Conversely, suppose a previous prefix channel r is dominated by an attainable s in $\mathcal{R}_{t-1}$. Conditional independence of the next observation implies that $E_t(s)$ dominates $E_t(r)$ by applying the same postprocessing to the state coordinate. Composing this postprocessing with any proposed new summary gives a binary quantizer of $E_t(s)$. The preceding argument dominates its output by a receiver-curve point, which belongs to $\mathcal{R}_t$. This composition can be incorporated in one update table and introduces no retained selector. At a one-state cut every prefix is uninformative and the reset to $(0, 0)$ is exact. This proves both inductive assertions.

A terminal readout can compose any dominating channel's postprocessing, so Γ_G is monotone under domination. The original finite machine class has a maximizer by compactness of its row simplices. The inductive domination statement supplies an attainable member of $\mathcal{R}_n$ with at least that value. No attainable member can exceed the optimum. This proves the maximum and attainment in (3.10), without needing a convexity claim about $\mathcal{R}_n$.

For the last formula, exchange labels to arrange $r_1 \geq r_0$. If $r_0 + r_1 \leq 1$, accept hypothesis one on state one and randomize on state zero with probability $(1 - r_0 - r_1)/(2 - r_0 - r_1)$. The two successes then equal $(1 - r_0)/(2 - r_0 - r_1)$. If $r_0 + r_1 \geq 1$, reject on state zero and accept on state one with probability $1/(r_0 + r_1)$; both successes equal $r_1/(r_0 + r_1)$. These are likelihood-ratio tests and are minimax by the same equal-error supporting-prior argument as for a finite receiver curve. Both expressions equal (3.11); uninformative endpoints give $1/2$ directly. Choosing the attaining prefix quantizers proves the final implementation assertion. $\square$

Equation (3.9) is a genuine minimax whole-schedule characterization for arbitrary finite horizons in this binary-register class. It uses the ordinary comparison order of binary experiments, not a minimax interchange for complete machines. The restriction to uncontrolled, conditionally independent reports is used exactly when the dominating prefix is tensored with the next report. General controlled experiments and larger minimax registers use the occupation theorem instead. The three-report result that follows is a closed-form evaluation, with a strict mixing gap, of this same noisy resource problem.

4. AN EXACT NOISY FRONTIER AND THE COST OF HIDDEN MIXING

Let $\theta \in \{0, 1\}$, and let Y_1, Y_2, Y_3 be conditionally independent binary reports with

$$\mathbb{P}_\theta(Y_t \neq \theta) = p, \quad 0 < p < 1/2.$$

Every report has overlapping support under the two hypotheses. A clocked machine has at most K_t labels after report t and makes a binary terminal decision. Updates are atomic stochastic operations; additional random-bit cuts are discussed below. Let $B_{\mathbf{K}}(p)$ be its optimal success under the fair prior, and let $V_{\mathbf{K}}(p)$ be its optimal minimum of the two conditional success probabilities. Both optimize over all stochastic machines of this profile. Put

$$d(p) = p(1 - p)(1 - 2p) > 0.$$

Theorem 4.1 (Complete three-report profile frontier). *For every positive integer profile (K_1, K_2, K_3) ,*

<i>profile condition</i>	$B_{\mathbf{K}}(p)$	$V_{\mathbf{K}}(p)$	
$K_3 = 1$	$1/2$	$1/2$	(4.1)
$K_3 \geq 2, \min(K_1, K_2) = 1$	$1 - p$	$1 - p$	
$K_1, K_3 \geq 2, K_2 = 2$	$1 - p + \frac{1}{2}d(p)$	$1 - p + \frac{1}{3}d(p)$	
$K_1, K_3 \geq 2, K_2 \geq 3$	$1 - p + d(p)$	$1 - p + d(p)$	

In the third line a minimax optimal machine retains Y_1 , then retains $Y_1 Y_2$ (the Boolean AND), and after the third report decides one with probability

$$\frac{(M_2, Y_3)}{\mathbb{P}(\hat{\theta} = 1 \mid M_2, Y_3)} \mid \begin{array}{cccc} (0, 0) & (0, 1) & (1, 0) & (1, 1) \\ 0 & 2/3 & 1 & 1. \end{array} \quad (4.2)$$

The two conditional success probabilities are equal. In particular, for $p = 1/4$ and profile $(2, 2, 2)$ the exact Bayesian and minimax values are $51/64$ and $25/32$, whereas profile $(2, 3, 2)$ has value $27/32$ for both objectives.

We prove the minimax assertion without passing to a convex hull of machines. The following elementary binary-channel domination is useful.

Lemma 4.2 (Reduction of a two-report binary register). *Every stochastic binary summary of (Y_1, Y_2) is, up to exchanging its two output labels, a stochastic postprocessing of a summary with*

$$\mathbb{P}(M = 1 \mid 00) = 0, \quad \mathbb{P}(M = 1 \mid 01) = \mathbb{P}(M = 1 \mid 10) = w, \quad \mathbb{P}(M = 1 \mid 11) = 1 \quad (4.3)$$

for some $w \in [0, 1]$. This domination holds simultaneously under both hypotheses and continues to hold after appending an independent third report.

Proof. Write u, v for the summary probabilities on 00, 11, and write b for the average of its probabilities on 01, 10. Those two words have the same probability under either hypothesis, so their average contains all relevant information. Exchange output labels if necessary to arrange $v \geq u$, and put $\alpha = v - u \in [0, 1]$ and $c = 2p(1 - p) \in (0, 1)$. Its output-one probabilities satisfy

$$r_0 = u + c(b - u) + p^2\alpha, \quad r_1 - r_0 = (1 - 2p)\alpha.$$

For the summary (4.3) these probabilities are $s_0 = p^2 + cw$, $s_1 = (1 - p)^2 + cw$. Seek a postprocessing that sends input zero to one with probability a and input one to one with probability $a + \alpha$. It is stochastic precisely when $0 \leq a \leq 1 - \alpha$, and it reproduces r_0, r_1 if

$$a = u + c(b - u) - c\alpha w.$$

As w ranges over $[0, 1]$, the right side ranges from $u + c(b - v)$ to $u + c(b - u)$. The upper endpoint is nonnegative. The lower endpoint is at most $1 - \alpha$, since

$$1 - \alpha - u - c(b - v) = (1 - v) + c(v - b) \geq (1 - v)(1 - c) \geq 0.$$

Consequently this interval meets $[0, 1 - \alpha]$. This proves the claim, including $\alpha = 0$. Applying the postprocessing independently of the third report proves the last statement. $\square$

Proof of Theorem 4.1. If the last register has one state, its terminal decision is independent of the reports. The fair-prior value is one half, and a fair terminal randomization gives minimax value one half. If an earlier register has one state, at most the last two reports remain informative. The unrestricted optimal success from two equally reliable reports is $1 - p$ under either criterion: use their common value on agreement and a fair decision on disagreement. One report already gives $1 - p$. These rules respect the stated profiles and prove the second line.

Suppose henceforth that $K_1, K_3 \geq 2$. There is no loss in retaining Y_1 exactly. Indeed any stochastic first update followed by a second update induces one binary channel from (Y_1, Y_2) to M_2 , and that channel can be implemented directly when Y_1 is retained. Allowing more than two first-stage labels gives no larger set of such channels. At the last stage any randomized binary decision based on (M_2, Y_3) can be placed directly in a two-label register. Thus, when $K_2 = 2$, the unrestricted optimization is exactly over a binary summary of the first two reports and a randomized readout of it together with Y_3 .

For the Bayesian problem, Theorem 3.2 reduces the middle update to intervals of the three distinct likelihood-ratio classes 00, $\{01, 10\}$, 11. A nonconstant two-interval update isolates 00 or

isolates 11. These are the Boolean OR and AND, up to labels. With AND, the Bayesian readout is $\hat{\theta} = Y_3 \vee (Y_1 \wedge Y_2)$. Its conditional successes are

$$c_0 = 1 - p - p^2 + p^3, \quad c_1 = 1 - 2p^2 + p^3.$$

Their average is $1 - p + d(p)/2$. The complementary rule interchanges c_0, c_1 . A constant middle update gives only $1 - p$, proving the Bayesian third line. This also covers updates that split the tied middle words: the theorem permits assigning the entire tied class to one maximizing suffix label without decreasing the Bayesian objective.

For the minimax problem, Lemma 4.2 reduces the search, without loss, to (4.3). Complementing all reports and output labels replaces w by $1 - w$ and interchanges the hypotheses, so it suffices to take $0 \leq w \leq 1/2$. Write

$$r_0 = p^2 + 2p(1 - p)w, \quad r_1 = (1 - p)^2 + 2p(1 - p)w.$$

In the experiment (M_2, Y_3) the likelihood ratios are ordered as 00, 01, 10, 11. Only the ordering of the middle two needs verification; after multiplying by their positive denominators it is equivalent to

$$r_1(1 - r_0)p^2 - r_0(1 - r_1)(1 - p)^2 = p^2(1 - p)^2(1 - 2p)(1 - 2w)^2 \geq 0.$$

Consider the test that accepts hypothesis one on 10, 11, rejects it on 00, and accepts it on 01 with probability

$$a(w) = \frac{1 - r_0 - r_1}{(1 - r_0)p + (1 - r_1)(1 - p)} = \frac{2(1 - 2w)}{3 - 2w}. \quad (4.4)$$

This number is in $[0, 1]$. Its two conditional success probabilities are equal, with common value

$$R(w) = 1 - r_0 - a(w)(1 - r_0)p. \quad (4.5)$$

It is a likelihood-ratio test, possibly randomized at one threshold. Choose the prior odds equal to that threshold. The test is then Bayesian optimal for those odds. Since its two successes are equal, no test can have a larger minimum: such a test would also have a larger payoff for this prior. Thus $R(w)$ is the exact minimax success for this summary, not merely an equal-error attainable value.

Direct differentiation gives

$$R'(w) = -\frac{2p(1 - p)(1 - 2p)(5 - 2w)(1 - 2w)}{(3 - 2w)^2} \leq 0. \quad (4.6)$$

Equivalently, an explicit nonnegative gap certificate is

$$R(0) - R(w) = \frac{2p(1 - p)(1 - 2p)w(5 - 6w)}{3(3 - 2w)} \geq 0, \quad 0 \leq w \leq 1/2. \quad (4.7)$$

The denominator is positive and every factor in the numerator is nonnegative on the stated domain. Therefore the best w is zero. Equations (4.4)–(4.5) give $a(0) = 2/3$ and

$$R(0) = 1 - \frac{2}{3}p - p^2 + \frac{2}{3}p^3 = 1 - p + \frac{1}{3}d(p).$$

The postprocessing in Lemma 4.2 can be composed with any final readout and does not need a retained selector. It follows that this upper bound applies to every original stochastic machine. The rule (4.2) attains it.

Finally, if $K_2 \geq 3$, retain the number of ones in the first two reports and output the majority of all three. Both successes are $1 - 3p^2 + 2p^3 = 1 - p + d(p)$. The likelihood-ratio rule for the full three reports is majority under the fair prior, so no machine can have a larger Bayesian value or a larger minimum. This proves the last line and completes all cases. $\square$

Corollary 4.3 (A strict fixed-profile mixing gap). *For profile $(2, 2, 2)$ a free initial selector that mixes the two complementary Bayesian optimal machines would give worst-case success $1 - p + d(p)/2$. Without that uncharged selector the exact value is $1 - p + d(p)/3$. The excess $d(p)/6$ is strictly positive.*

Proof. Mix the two payoff vectors (c_0, c_1) and (c_1, c_0) from the proof with equal weights. Each coordinate becomes their average. Theorem 4.1 gives the smaller all-stochastic value at the original profile. To implement the mixture literally, retain the selector jointly with the machine state; a profile $(4, 4, 2)$ suffices. The corollary does not claim that this product construction minimizes the expanded profile. $\square$

The gap identifies a specific failure of a convexified all-machine converse. A terminal output law can be a convex combination of laws while the machines producing those laws cannot be mixed within their original memory profile. In contrast, the quadratic compatibility constraints of Theorem 2.1 retain exactly the required coupling. The noisy loss $2d(p)/3$ between two and three middle states is also a strict full-schedule minimax loss; it is not an erasure probability or the cost of remembering a label that was already known.

Remark 4.4 (Atomic randomization and finite precision). The fraction $2/3$ in (4.2) is an exact rational row of the all-stochastic model. A fixed finite string of fair bits cannot realize this fraction exactly. Replacing it by a dyadic number within 2^{-r} gives a genuine finite-precision rule whose minimum success is at least $V_{(2,2,2)}(p) - 2^{-r}$. If each random-bit epoch is a charged cut, retain (M_2, Y_3) while executing the dyadic sampler; the three-status comparison sampler of Section 10 gives a conservative peak of 12 for this refined execution. The theorem’s exact two-state value refers to its stated atomic-update profile, not to this finer profile. The latter has its own exact occupation formulation. Neither an unbounded rejection sampler nor a free fair-bit clock is hidden in the two-state assertion.

5. ACQUISITION, CLOCKS, AND THE RESOURCE ORDER

A statistical bound is ambiguous unless the protocol and the resource coordinates being compared are specified. The superscript clk in the preceding theorems is therefore part of their notation, not a qualification confined to an implementation paragraph.

5.1. Comparable resource points. On one declared schedule, profiles are ordered coordinatewise. Enlarging a register alphabet preserves every previously feasible machine by padding unused states. Inserting observations is comparable only through an explicit embedding that says which new inputs are ignored and which registers survive the inserted cuts. Atomic and bit-serial schedules are different schedules, even when they have the same number of experimental preparations.

For a finite-precision implementation we record

$$\mathbf{r} = (N_B, N_Q, \mathbf{K}, r, D, \mathbf{C}). \quad (5.1)$$

Here N_B, N_Q count candidate and target preparations, r bounds fair bits per declared row sampler, D bounds a specified read-only description, and $\mathbf{C}$ specifies clocked or autonomous execution. A description is measured in a fixed code, not by an unspecified notion of compressibility. For example, with a known table layout, store the first $s - 1$ numerators of a 2^{-r} stochastic row in $(s - 1)(r + 1)$ bits; its last numerator is the remainder. The layout, response dictionary and any program used to expand repeated phases have their own fixed-code costs. Sample-dependent calibration is not part of that read-only description.

For the same code and schedule, increasing numerical resource caps or register alphabets gives an inclusion of machine classes. Changing $\mathbf{C}$ does not mean increasing one numerical coordinate. A clocked execution with T update epochs and profile $\mathbf{K}$ has an autonomous realization on the

disjoint union of its phase registers, using at most $\sum_{t=0}^T K_t$ states. This is an attainable conversion, not an equality of optimal frontiers. Shared-row optimization in Theorem 2.1 can be strictly smaller than that conversion.

Proposition 5.1 (Joint finite resource optimization). *Fix a finite experimental protocol and a finite table code. With finite caps on description and precision, the feasible machines form a finite set and their optimal worst-case payoff is exactly the maximum over that set. If the precision restriction is removed on a fixed table architecture, Theorems 2.1 and 2.2 characterize its exact value. A finite collection of permitted architectures is handled by taking the maximum of their values. Calibration calls can be included as an initial part of the same protocol, with all their retained information passed through the same profile constraints.*

Proof. There are finitely many descriptions of bounded length and finitely many allowed dyadic rows. Retain exactly the descriptions that decode to legal tables of the specified finite schedule and budgets. Each has a well-defined payoff. For unrestricted stochastic rows on a fixed architecture, use its observable row sets in the occupation formulation. Optimizing over a finite union is a maximum over its components. A calibration prefix uses known experimental kernels at the unknown environment in exactly the same way as any other prefix; its sample-dependent state is an ordinary retained register. Thus it introduces neither a parameter oracle nor a new exception to the compatibility constraints. $\square$

The proposition supplies a joint optimization model rather than merely a post hoc resource ledger. It does not claim a short closed formula for arbitrary description budgets. For compact continuous families the net condition in Corollary 2.3 is still required for effective finite certificates.

5.2. Acquiring a calibration rather than assuming it. The next result gives a fully priced constructive calibration prefix. Let $h_1, \dots, h_K : \Omega \rightarrow [0, 1]$ be a fixed known response list, chosen independently of the calibration observations, and suppose its values have denominator 2^r . Write $d = |\Omega|$ and $q_i = Qh_i$. The target Q need not be given as a table: independent preparations from Q are available. Let $M = 2^s$ and $0 < \beta < 1$.

Theorem 5.2 (Sampled calibration with persistent-register pricing). *For every $\rho > 0$ with*

$$M \geq \frac{1}{2\rho^2} \log \frac{2K}{\beta}, \quad (5.2)$$

there is a clocked finite-precision calibration protocol using M target preparations and MKr fair bits. It returns $\hat{q}_i \in \{0, 1/M, \dots, 1\}$ with

$$\mathbb{P} \left(\max_i |\hat{q}_i - q_i| \leq \rho \right) \geq 1 - \beta.$$

The complete counter register has at most $C = (M + 1)^K$ states; a word buffer and the three-status dyadic sampler give peak at most $3dC$. If a subsequent auditor has peak at most W for any exogenously fixed calibration vector, retaining the acquired calibration gives the bound $C \max(3d, W)$ for the combined execution.

Let $v \in [-1, 1]$ and $\epsilon \geq 0$. Suppose, uniformly for calibration vectors $\tilde{q}$, the auditor's true expected score is at least $v - \epsilon - 2\|\tilde{q} - q\|_\infty$, and its score is in $[-1, 1]$. With fresh training and validation after this prefix, the combined expected score is at least

$$v - \epsilon - 2\rho - 2\beta. \quad (5.3)$$

Proof. On target trial j , retain its word Y_j . For each i , generate a Bernoulli variable B_{ij} of parameter $h_i(Y_j)$ with an independent r -bit dyadic sampler, and increment counter i . After all

K updates, discard Y_j . For fixed i , the variables B_{ij} over different trials are independent, lie in $\{0, 1\}$, and have mean q_i . Their averages $\hat{q}_i$ therefore satisfy

$$\mathbb{P}(|\hat{q}_i - q_i| > \rho) \leq 2e^{-2M\rho^2}.$$

This bounded-variable exponential estimate follows from the exponential moment bound used in Theorem 9.2. A union bound proves the probability assertion. Independence between different counters is not required.

The K counters each have $M + 1$ possible values. The word buffer has d values, and the running comparison of a uniform r -bit word with a fixed dyadic threshold has three comparison statuses. These yield the stated state counts at every sampler cut. Trial, response and bit indices are in the declared clocked schedule, not in an unmentioned work tape. The persistent counter vector remains in the register of the subsequent auditor and multiplies its width by at most C . All its values are dyadic because $M = 2^s$.

Conditional on the calibration vector, later observations have their original independent laws. On the good calibration event the score bound is $v - \epsilon - 2\rho$. On the complementary event the score is at least -1 . Since $v \leq 1$ and $\epsilon, \rho \geq 0$, a loss of at most two times that event's probability suffices. This proves (5.3). $\square$

For the response-cover tournament of Theorem 13.2, one may take

$$N_B = n(K - 1), \quad W = 3dK(2m + 1), \quad \epsilon = 4(K - 1)\theta + 2(K - 1)(e^{-4\theta m} + e^{-2n\theta^2})$$

when v is the envelope of the already dyadic response list. Starting with an arbitrary list and rounding it to r bits incurs the additional 2^{1-r} term of that theorem. The new prefix prices target acquisition and its persistent calibration state. It is an explicit construction, not an optimal calibration-memory claim. Joint optimality at fixed finite budgets is the distinct problem in Proposition 5.1.

6. CONSUMERS OF THE COMPATIBLE FRONTIER

The preceding characterizations and the constructive continuation results below have the same object of optimization: one experiment, one schedule and one whole register profile. This section records the connections before developing the underlying marked and physical experiments in detail.

6.1. Reset scores and physical protocols. In the reset interface of Section 9, a fixed candidate supplies the training and candidate-validation kernels. The target supplies its own preparation and validation kernels. The final score is a function of the two retained validation bits. Include the response selection, all input-bit buffers, and all randomization cuts in σ . This produces precisely the finite protocol of Section 2. The image of the candidate family is not replaced by its convex hull.

Corollary 6.1 (Common constructive and converse interface). *At any fixed finite reset protocol and full profile, the best worst-case score has the occupation characterization of Theorem 2.1; for a finite family it has the complete dual of Theorem 2.2. For a compact family with effective finite kernel nets it has the converging upper and lower certificates of Corollary 2.3. The response-cover machines of Theorems 12.2 and 13.2 supply feasible lower certificates in these very classes. Any certified dual bound applies to those machines and to every other stochastic machine with the same protocol and profile.*

Proof. Augment the hidden index by the fixed candidate parameter and the finite public trial history. Selection and memory updates are parameter-independent observer rows. Experimental draws are the known linear environment operations. Terminal scoring is linear in the terminal

occupation array. Thus the realization theorem applies in both directions. The constructive machines specify legal rows at every cut and hence feasible occupation arrays. Compact-family and dual assertions are exactly the corresponding general theorems. $\square$

For the positive-noise marked collision experiment, the finite row-report alphabet is fixed and the private row family is compact. On compact subfamilies with the stated continuous kernel data, finite kernel nets give the preceding certificates. The microscopic and Gaussian comparison theorems below then transfer a certified payoff with their stated common-preparation error. If training kernels differ, their errors must also be charged; equality of the training source cannot be inferred from equality of the terminal validation task. The active-state estimate in Section 15 can replace a worst-case telescoping estimate for a fixed retained execution, but is not a bound on a full discarded transcript.

Thus the physical resource constructions (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ are now accompanied by a general all-auditor converse procedure on each fixed full profile. They remain attainable resource points, not a list of proved Pareto optima. In particular, the numerical statement $3 \leq W_{2/5}^{\text{phys}} \leq 12$ is not changed into an equality without an additional certificate or construction.

6.2. An exact confidence consumer. Let a reset audit return $D \in \{-1, 0, 1\}$. Suppose its null mean is zero and its alternative mean is at least g , where $0 < g \leq 1$ is known. On each independent repetition, use a fresh coin with

$$\mathbb{P}(Z = 1 \mid D) = \frac{1}{2} + \frac{D - g/2}{4}. \quad (6.1)$$

Under the null its success probability is $p = 1/2 - g/8$, and under the alternative it is at least $1 - p$. Consider the declared coin interface consisting of the null Bernoulli law p and all alternative Bernoulli laws with parameter in $[1 - p, 1]$, independently repeated.

Corollary 6.2 (Exact three-repetition confidence frontier). *At the atomic retained-coin profile (2, 2, 2), the minimum possible maximum of type-I and type-II error over this coin interface is*

$$p - \frac{1}{3}p(1 - p)(1 - 2p). \quad (6.2)$$

The rule (4.2) attains it. Increasing only the middle retained-coin alphabet to three changes this value to $p - p(1 - p)(1 - 2p)$. If one underlying audit uses b candidate training trials, two validation trials, and peak W , the three-repetition construction uses $3(b + 2)$ preparations. Its inner audit, coin sampler, and outer state are charged jointly, rather than identifying an outer two-state register with the entire physical register.

Proof. The alternative family contains the endpoint $1 - p$, so Theorem 4.1 gives the lower bound on the maximum error. For the attaining test, its probability of deciding one, as a function of the three input bits, is coordinatewise nondecreasing: the middle state is AND, and the four final probabilities are 0, $2/3$, 1, 1. Couple Bernoulli inputs with common independent uniforms. The power is then nondecreasing in their parameter, so the largest type-II error occurs at the endpoint. This proves attainment for the whole declared alternative interval. The majority test proves the assertion with three middle states in the same way. Preparation counts follow by repeating the base experiment independently three times. Product registers implement the outer transitions together with each inner repetition. $\square$

If g is dyadic, the coin in (6.1) has a finite dyadic sampler. At its internal cuts an explicit conservative bound is $2 \max(W, 9)$ for the outer state and inner audit/coin sampler; the additional $2/3$ terminal row is either an atomic rational operation or is replaced by the finite-precision realization

in Remark 4.4. A fully fair-bit-serial realization has the conservative peak $\max\{2\max(W, 9), 12\}$ and the indicated 2^{-r} error allowance. This does not erase its clock or its bits from the schedule.

The exact lower bound concerns the declared coin interface. A physical subfamily inherits that lower bound only if it contains its endpoint experiment (or an appropriate certified limiting experiment). A raw physical auditor that uses more of D or of the original report is not restricted to this interface. For such auditors the general occupation converse, with those richer inputs retained, is the relevant statement. The confidence result is therefore a proved consumer, not a relabeling of an expected-score lower bound as a hypothesis-testing lower bound.

6.3. The foundation and the longer program. The initial objects remain positive experimental kernels, preparations, causal interventions and executable continuations. Finite protocols are a resource-controlled subcategory, not a replacement for the standard Borel formulation or its limiting targets. The new occupation and continuation results supply a precise G3-type finite resource layer. The marked deficiency and Gaussian transport results supply the comparison maps used above. The finite resource theorems do not supply a missing kinetic generator domain, a compactness theorem for collision measures, or an optional-projection theorem on a general operator domain.

The full historical dependency graph is retained in Appendix B. The geometric A2 chain has its own premises. The B4 state-dependent control-transfer, range and compactness targets, and the broader C2 operator and rigidity targets, retain their original scope. Their relation to this paper is through explicit experiment and simulator interfaces when those have been established, not through a declaration that every manuscript is a consequence of a finite-state comparison theorem.

7. INTRINSIC CAUSAL DEFICIENCY

7.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \tag{7.1}$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \tag{7.2}$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta, g}^{q^u} - Q_{\theta, g} \right\|_{\text{TV}}, \quad (7.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (7.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed–marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 7.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

7.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta, g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta, g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (7.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff} \{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta, g}(B) - P_{\theta, g}^d(B). \quad (7.5)$$

Including both an event and its complement includes both signs.

Theorem 7.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min \{ \varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta, g}(B) - P_{\theta, g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I} \}. \quad (7.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (7.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (7.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (7.8)$$

(iii) Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (7.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (7.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (7.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) For compatible experiments E, F, G and profiles K, L ,

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (7.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multiaffine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (7.2); the same proof includes λ and the r policies in (7.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (7.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (7.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (7.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives

coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (7.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (7.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (7.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (7.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (7.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 7.3 (What the converse says). The dual (7.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

7.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 7.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (7.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (7.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1-\varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1-\varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (7.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (7.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (7.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 7.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

7.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (7.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 7.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 7.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 7.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 7.6 (Finite computation and general carriers). For rational finite instruments, (7.6) and the fixed- r formulations are finite polynomial feasibility problems; (7.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

8. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

8.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the

current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 8.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (8.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 8.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (8.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (8.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 7.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 7.2 therefore continues to give equality of the visible and private zero sets.

8.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target's terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 8.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (8.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m | x)$ and $d(z | m)$. Its joint terminal law is $\pi_x \sum_m e(m | x) d(z | m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x | m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

9. DECISION CONTINUATION SPECTRA

The memory at an experimental cut has a different role from the memory needed to reach that cut. We first identify its exact asymptotic value. This will give both an obstruction and

a construction for the same physical audit, rather than a lower bound for an auxiliary binary problem.

9.1. A charged training-validation interface. Let Ω be a finite marked alphabet, $d = |\Omega|$, let $B \subset \Delta(\Omega)$ be nonempty and compact, and fix $Q \in \Delta(\Omega)$. A candidate $b \in B$ is fixed throughout the experiment. Training consists of n independent observations from b . At the end of training the auditor retains a register $M \in [K]$. Everything depending on the training observations or on an earlier random seed must pass through this register. In particular, there is no external copy of the training history and no uncharged persistent selector.

After this cut, fresh independent randomness may select an event A using M . The event is chosen before independent candidate and target validation observations $X \sim b$ and $Y \sim Q$ are drawn. The score is $\mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The downstream register needed to keep A and to compute this score is distinct from the register at the specified cut. The upstream and downstream widths will be stated separately in concrete implementations. This is the reset interface of Section 18, with one additional charged checkpoint. The actual mark is part of a completed observation; it is not an input to the private candidate's online transition rows.

Write $\mathcal{H} = [0, 1]^\Omega$ and

$$f_h(b) = \sum_{w \in \Omega} (Q(w) - b(w))h(w), \quad h \in \mathcal{H}.$$

Every randomized event has a response $h(w) = \Pr(w \in A)$. Conversely, every $h \in \mathcal{H}$ is the response of a random event: include each atom independently with its specified probability. This is a finite probability distribution on 2^Ω . Only its mean response matters for expected signed score. The event itself must still be retained during validation.

Definition 9.1 (Decision continuation spectrum). The K -state decision continuation value is

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K \in \mathcal{H}} \min_{b \in B} \max_{m \leq K} f_{h_m}(b). \quad (9.1)$$

Let $v_{n,K}(B, Q)$ be the supremum of the worst-case expected score of n -sample auditors at the preceding interface. Only the designated cut has width K in this definition; all other widths are finite but unrestricted. For $c \in \mathbb{R}$, let $\kappa_c(B, Q)$ be the least K for which $\mathfrak{V}_K(B, Q) > c$, with value $+\infty$ when no such K exists.

The definition uses only the marked behavior image and its target, not a parameterization of B or a labeling of candidate states. No convex hull of B is taken. A response list is a list of conditional validation continuations; it is not a mixture of feasible private candidates.

Theorem 9.2 (Operational classification at a decision cut). *All extrema in (9.1) are attained. For every $n \geq 1$,*

$$0 \leq \mathfrak{V}_K(B, Q) - v_{n,K}(B, Q) \leq \min \left\{ \sqrt{\frac{d-1}{n}}, \sqrt{\frac{2 \log(2K)}{n}} \right\}. \quad (9.2)$$

Consequently $v_{n,K} \rightarrow \mathfrak{V}_K$. Every auditor with K states at the designated cut, at any finite training length, has worst-case score at most $\mathfrak{V}_K$. If $\mathfrak{V}_K > c$, some finite-sample auditor with K states at that cut has worst-case score greater than c . Thus κ_c is exactly the minimum cut width permitting such a strict certificate with sufficiently many training samples.

Equivalently, κ_c is the least number of response half-spaces

$$\{b \in B : f_h(b) > c\}, \quad h \in \mathcal{H}, \quad (9.3)$$

that cover B . If two parameterized behavior families satisfy, uniformly in their common parameter q ,

$$\|b(q) - \tilde{b}(q)\|_{\text{TV}} \leq \alpha, \quad \|Q - \tilde{Q}\|_{\text{TV}} \leq \beta,$$

then

$$|\mathfrak{V}_K(B, Q) - \mathfrak{V}_K(\tilde{B}, \tilde{Q})| \leq \alpha + \beta. \quad (9.4)$$

Proof. For a fixed response list, the inner objective is continuous in b . Its variation when the list changes is bounded uniformly over b by $2 \max_m \|h_m - \tilde{h}_m\|_\infty$. Its minimum over compact B is therefore continuous in the list. Compactness of $\mathcal{H}^K$ proves attainment.

Condition on the retained state of any admissible auditor. Its conditional post-cut event law depends only on that state and fresh randomness, and has a response h_m . Independence of validation from training gives

$$g(b) = \sum_{m=1}^K \Pr_b(M = m) f_{h_m}(b) \leq \max_m f_{h_m}(b).$$

Taking the minimum over b , and then the supremum over auditors, proves the upper bound. This argument permits arbitrary stochastic training and arbitrary finite upstream memory. An extra training-dependent decoder would invalidate its first conditioning step and is excluded by the interface, not silently supplied.

For the reverse bound fix an optimal list. Let $\hat{b}$ be the empirical measure and choose the first index maximizing $f_{h_m}(\hat{b})$. Retain that index and discard the empirical table. If m_* maximizes the true score, then pointwise

$$\max_m f_{h_m}(b) - f_{h_{m_*}}(b) \leq 2 \max_{m \leq K} |(\hat{b} - b)h_m|. \quad (9.5)$$

The maximum is at most $\|\hat{b} - b\|_{\text{TV}}$, since $0 \leq h_m \leq 1$. Cauchy–Schwarz and the multinomial variances give

$$\mathbb{E}\|\hat{b} - b\|_{\text{TV}}^2 \leq \frac{d}{4n} \left(1 - \sum_w b(w)^2\right) \leq \frac{d-1}{4n}.$$

This proves the first rate.

For completeness, if V takes values in an interval of length one, then $\log \mathbb{E}e^{s(V - \mathbb{E}V)} \leq s^2/8$. Indeed the second derivative of the logarithmic moment generating function is the variance under the exponentially tilted law, at most $1/4$; its value and first derivative vanish at zero. Integrating twice proves the inequality for both signs of s . Independence consequently gives $\mathbb{E}e^{s(b-b)h_m} \leq e^{s^2/(8n)}$. For $s > 0$,

$$\mathbb{E} \max_m |(\hat{b} - b)h_m| \leq \frac{\log(2K)}{s} + \frac{s}{8n}.$$

Minimizing in s gives $\sqrt{\log(2K)/(2n)}$ and proves the second rate in (9.2). The empirical table has finitely many values; the construction is therefore a finite-state auditor, with only the chosen index surviving the specified cut. Real response coefficients are allowed here; the physical construction below has deterministic rational transitions.

A finite cover in (9.3) says that the continuous function $b \mapsto \max_m f_{h_m}(b)$ is greater than c everywhere on compact B . Its attained minimum is then greater than c . This proves the covering characterization in both directions. It also proves the strict-threshold assertion; no claim about finite-sample attainment at the boundary $c = \mathfrak{V}_K$ is needed.

Finally $|(Q - b)h - (\tilde{Q} - \tilde{b})h| \leq \alpha + \beta$ for every response. Taking maxima, minima over the common parameter set, and maxima over lists preserves this uniform bound, proving (9.4). $\square$

Corollary 9.3 (Compression by a finite decision dictionary). *Suppose responses $h_1, \dots, h_K$ satisfy $\min_{b \in B} \max_m f_{h_m}(b) \geq c + \eta$ for $\eta > 0$. An empirical selector with $n > 2 \log(2K)/\eta^2$ achieves score greater than c and retains only K states at the decision cut. If these responses depend on a finite statistic $T : \Omega \rightarrow \Xi$, the training selector can count $T(X)$ instead of X . A sufficient*

upstream register has at most $\binom{n+|\Xi|-1}{|\Xi|-1}$ states at the last training checkpoint, and the corresponding time-dependent count registers suffice at earlier checkpoints.

Proof. Use the second bound in (9.2). Empirical expectations of h_m depend only on the histogram of $T(X)$. At time t , that histogram has $\binom{t+|\Xi|-1}{|\Xi|-1}$ possibilities. Its update and the final maximization are deterministic. No statistical sufficiency for estimating all of b is required; sufficiency for the specified responses is enough. $\square$

This corollary supplies an intermediate principle between a full marked histogram and a problem-specific compressed counter. The invariant is the number and geometry of validation continuations needed to cover the original candidate image at the desired score. It is not the algebraic dimension of a chosen parameterization. The empirical upper bound need not be sample-optimal, and a minimum decision-cut width is not a minimum of the largest width over the entire acquisition schedule.

9.2. Several feedback rows. The classification does not require that the declared experiment have only one feedback row. For the row-indexed interface of Section 18, let $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$ and define

$$f_\lambda(b) = \sum_{i \in \mathcal{I}} \lambda_i f_i(b), \quad \lambda \in \Delta(\mathcal{I}).$$

A post-cut continuation is now a probability distribution on row–event pairs; it is not an arbitrary test on a joint product of validation rows. Put

$$\mathfrak{V}_K^{\text{rows}}(B, Q) = \max_{\lambda_1, \dots, \lambda_K \in \Delta(\mathcal{I})} \min_{b \in B} \max_m f_{\lambda_m}(b).$$

Corollary 9.4 (Row-indexed classification). *With n training samples per row, the row-indexed decision-cut value converges to $\mathfrak{V}_K^{\text{rows}}$, and its deficit is at most*

$$\min\{\sqrt{D/n}, \sqrt{2 \log(2K)/n}\}, \quad D = \sum_j (|\Omega_j| - 1).$$

The strict half-space cover characterization holds with f_λ . The upper bound by $\mathfrak{V}_K^{\text{rows}}$ applies also to any other finite training schedule, including adaptive row selection.

Proof. Conditioning on the cut state gives the same convex-mixture upper bound as before; its proof places no restriction on how that state was reached. For empirical selection among an optimal list, $|(\hat{b} - b)\lambda_m| \leq \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Sum the rowwise second moments to obtain the first rate. To obtain the second, align the r th independent training sample from each row into a tuple and write its response as $\sum_{j,A} \lambda_{m,j,A} \mathbf{1}_A(X_{j,r})$. This random variable belongs to $[0, 1]$, and its sample mean is the empirical candidate mean for that continuation. The logarithmic moment-generating argument in Theorem 9.2 therefore gives the same finite-list rate. Each chosen row–event distribution can be drawn using fresh randomness after the cut. Compactness and the covering argument are unchanged. $\square$

10. FINITE PRECISION AND CHARGED RANDOMIZATION

We distinguish the number of persistent labels from the description of a transition and from the clock that selects it. An observation in a finite alphabet is initially an atomic input. Every random bit used after that input is a separate, charged epoch. Section 11 gives the additional cost of splitting the observation itself.

Lemma 10.1 (Dyadic rows and finite randomizers). *A probability vector on s points has a dyadic approximation with denominator 2^r at total variation distance at most $(s-1)2^{-r}$. It can be sampled using exactly r independent fair bits and at most $2s-1$ persistent labels at each intermediate bit*

epoch. The label count uses the bit position supplied by the schedule. No partial random integer is retained outside these labels.

Proof. Round down the first $s - 1$ coordinates and assign the remaining mass to the last coordinate. The positive and negative parts of the change have equal mass, at most $(s - 1)2^{-r}$. For sampling, partition $\{0, \dots, 2^r - 1\}$ into consecutive intervals of the prescribed integer lengths. A binary prefix describes a dyadic interval. If that interval is contained in one output interval, keep the output label and ignore the remaining bits. Otherwise keep the prefix interval. At a fixed depth, every unresolved interval contains a partition boundary in its interior; there are at most $s - 1$ such intervals. Together with at most s resolved outputs this gives $2s - 1$ labels. The children and their resolved labels are fixed by the read-only partition and the depth. Thus the construction is an actual finite transition table, not a real-valued random seed. Zero-length intervals cause no difficulty. $\square$

Proposition 10.2 (Rational decision continuations). *Let $d = |\Omega|$. A list of K responses in $[0, 1]^\Omega$ can be replaced by dyadic responses of denominator 2^r , losing at most 2^{1-r} in its uniform score envelope. After a decision cut retaining K labels, the randomized event for the chosen dyadic response can be drawn using r fair bits, at most $K(2d + 1)$ labels during this draw, and at most $K(d + 1)$ labels afterwards. Keeping the event through one candidate and one target validation requires at most $2K(d + 1)$ labels at the intervening atomic-observation cuts. Its final signed score has three values.*

Proof. Round every coordinate down. The sup-norm error is at most 2^{-r} , and $\sum_w |Q(w) - b(w)| \leq 2$. This proves the envelope estimate. For a fixed rounded response h , draw a uniform dyadic integer U and choose $A_U = \{w : U < 2^r h(w)\}$. These events are nested and have at most $d + 1$ distinct values. Their mean response is exactly h . Apply Lemma 10.1 to their interval lengths, retaining the chosen response index during sampling. Afterwards the event label and the candidate-event indicator suffice; the target indicator emits their difference. All event masks are read-only data fixed before observing the candidate. $\square$

For K responses, their rounded coordinates require $Kd(r + 1)$ bits in a direct read-only representation. These bits are not charged as persistent sample-dependent labels, but are not declared costless precision. A construction using them records both quantities. In particular, the proposition turns every strict response-cover margin into a finite-randomness implementation; it says nothing about exact attainment of an irrational boundary value.

11. APPROXIMATE CONTINUATION PROFILES AND RESOURCE TRANSFER

A cut profile records the whole causal schedule, rather than its final compression. We give both a realization certificate and an operational value. These notions keep the compatibility condition of Theorem 8.2; separate nonnegative factorizations at individual cuts are not a substitute for that condition.

11.1. A causal comparison norm. For two normalized finite causal suffix kernels p, q at cut t , put

$$d_t^{\text{ca}}(p, q) = \sup_{\pi} \|P_p^\pi - P_q^\pi\|_{\text{TV}},$$

where π ranges over deterministic causal input policies on the remaining schedule and P_p^π is the law of its complete output transcript. Randomized policies give the same supremum, by conditioning on their independent random tables. In particular this distance includes feedback; the supremum over open-loop input words alone is not used.

For stochastic coefficients $q_t(z, j \mid m, x)$ and suffix kernels $v_{t+1,j}$, denote by $\mathcal{Q}_{t+1,m}$ the suffix kernel which, upon input x , emits z and chooses j with these coefficients, then uses $v_{t+1,j}$. The notation denotes one first-step operation, not a conditioning on an event of zero probability.

Theorem 11.1 (Approximate compatible realization). *Let $K_0, \dots, K_L$ be positive integers. Suppose normalized causal kernels $v_{t,m}$, $m \leq K_t$, and stochastic rows q_t satisfy*

$$d_t^{\text{ca}}(v_{t,m}, \mathcal{Q}_t v_{t+1,m}) \leq \eta_t \quad (t < L, m \leq K_t). \quad (11.1)$$

At the terminal cut the suffix kernels are the unit empty kernel. If $\rho \in \Delta_{K_0}$ and $d_0^{\text{ca}}(p, \sum_m \rho_m v_{0,m}) \leq \eta_{-1}$, then the machine with initial law ρ and transition rows q_t has the whole profile (K_t) and realizes a kernel $\hat{p}$ satisfying

$$d_0^{\text{ca}}(p, \hat{p}) \leq \min\{1, \eta_{-1} + \sum_{t=0}^{L-1} \eta_t\}. \quad (11.2)$$

Conversely, a machine with this profile within distance ε of p supplies such a certificate with $\eta_{-1} = \varepsilon$ and all local errors zero. If a deterministic initial state is required, use $K_0 = 1$ throughout both statements.

Proof. Let $\hat{v}_{t,m}$ be the actual continuation of the rows q from state m . Fix a future policy. Conditional on the first output and next state, its remaining actions form an admissible suffix policy. Replacing $v_{t+1,j}$ by $\hat{v}_{t+1,j}$ therefore changes its transcript law by at most $\max_j d_{t+1}^{\text{ca}}(v_{t+1,j}, \hat{v}_{t+1,j})$. Mixtures and pushforwards contract total variation. The triangle inequality and (11.1) imply, backwards from L ,

$$\max_m d_t^{\text{ca}}(v_{t,m}, \hat{v}_{t,m}) \leq \sum_{s=t}^{L-1} \eta_s.$$

Mixing with ρ proves (11.2). The converse takes the actual normalized continuations of the given machine, including arbitrary valid continuations for unreachable states. Their one-step equations are identities. There is no assertion that independently optimal approximate lists at successive cuts have compatible updates. $\square$

11.2. The full value and its effective approximation. Fix a finite schedule with finitely many observation and action symbols, N calls to an experiment, and a bounded terminal payoff $G \in [-1, 1]$. An environment $b \in B$ specifies the conditional observation kernels at all possible visible histories and actions. Assume B is a nonempty compact subset of the resulting finite product of probability simplexes. A controller A has a specified persistent profile $\mathbf{K}$; the clock has no access to sample history. All action choices and all retained randomness factor through the current register. Define

$$V_{N,\mathbf{K}}(B, G) = \max_{A \in \mathcal{M}(\mathbf{K})} \min_{b \in B} \mathbb{E}_b^A G. \quad (11.3)$$

Unused states pad smaller machines, so $\mathcal{M}(\mathbf{K})$ is a finite product of stochastic-row simplexes, with any prescribed causal zeros imposed. The terminal readout is included among its rows.

Theorem 11.2 (Full-profile resource transfer). *The maximum in (11.3) is attained. Its value is nondecreasing in each K_t . Removing a charged intermediate cut gives a relaxation, whereas inserting identity-holding cuts preserves every implementation with the same held register.*

Suppose two environments at each of their N calls differ in conditional total variation by at most δ_i , uniformly over corresponding histories, actions and parameters. Replace the controller rows at its L transitions by dyadic rows of denominator 2^r , with at most s_t possible next-state/output pairs at transition t . Any fixed controller loses at most

$$2 \min \left\{ 1, \sum_{i=1}^N \delta_i + \sum_{t=1}^L (s_t - 1) 2^{-r} \right\} \quad (11.4)$$

in expected payoff. The same bound compares optimal values on the same profile, with the appropriate dyadic approximation on one side. More generally an approximate continuation certificate contributes its error in (11.2) inside the braces.

If $B_0 \subset B$ is a finite ω -net in the maximum conditional-TV metric and v_0 is the maximum over the dyadic controller grid of the minimum over B_0 , then

$$v_0 - 2N\omega \leq V_{N,\mathbf{K}}(B, G) \leq v_0 + 2 \sum_{t=1}^L (s_t - 1)2^{-r}. \quad (11.5)$$

The bounds are certificates whenever the net and the finite evaluations are certified. Compactness alone supplies existence of nets, not an algorithm for an arbitrary presentation of B .

Proof. For fixed b , each path probability is a finite product of controller and environment coordinates. Summing the finitely many paths gives a jointly continuous payoff. Its minimum over compact B is continuous in the controller rows; the controller space is compact. This proves attainment. Padding proves monotonicity. Removing a constraint enlarges the feasible set, and identity transitions prove the insertion assertion.

For the perturbation bound, couple two executions until their first mismatch. On equal histories the conditional observation laws can be maximally coupled, with mismatch probability at most δ_i . Couple the controller transitions similarly, using Lemma 10.1. The probability of any mismatch is at most the sum in (11.4), capped at one. The payoff changes by at most two on that event. This coupling is recursive on finite histories and remains valid for feedback controls. Alternatively, approximate continuation kernels are compared in the causal distance and then composed, as in Theorem 11.1.

For a fixed grid controller, a minimizing point of B has a point of B_0 within ω ; the corresponding payoffs differ by at most $2N\omega$. Thus the grid optimum on all of B is at least $v_0 - 2N\omega$. Conversely approximate a maximizing unrestricted controller row by row. Its minimum over B decreases by at most the row-rounding bound, and its minimum over B_0 is no smaller. Taking the grid maximum proves the upper bound. No minimax interchange or convexification of the fixed-width behavior image occurs. $\square$

The formula records preparation count, persistent labels, calibration error, causal simulation error and transition precision in one execution. The training-count factor is present when training kernels change. It is absent only when those kernels are identical. Failure probabilities are also transported, with factor one rather than two, since their indicators have range $[0, 1]$.

Proposition 11.3 (Composition and schedule refinement). *A causal simulator of profile $\mathbf{R}$ followed on the same synchronized schedule by a controller of profile $\mathbf{K}$ has a realization with profile $(R_t K_t)$. Causal simulation errors add under successive composition. Two disjoint phases with a genuine reset and no retained bridge use the maximum of their peak widths. If a bridge with H labels must survive the second phase, its direct width bound is H times the second-phase width.*

If one atomic input x is encoded by ℓ symbols, let D_j be the number of possible length- j prefixes. Holding the pre-input state and the prefix realizes the refined acquisition cuts with widths $K_{\text{before}} D_j$, followed by the original update. A time-dependent machine running for T scheduled epochs becomes autonomous by retaining its clock, with at most $(T + 1) \max_t K_t$ states. These are implementation bounds, not claims of optimality for every refinement.

Proof. Keep the ordered pair of simulator and controller registers and compose their transition kernels. The simulator's emitted observation is the controller's current input. Conditional coupling until the first simulation discrepancy gives the error statement. A genuine reset permits reusing the alphabet, whereas a surviving bridge is another register. A partial encoded input is determined by its prefix, giving the stated refinement construction. Finally the augmented state (t, m) implements the time-indexed transition without an external clock. These constructions also show why neither hidden selectors nor an external transcript may be omitted from the register being counted. $\square$

12. FROM A DECISION COVER TO AN END-TO-END MACHINE

The unrestricted empirical table in Theorem 9.2 can be replaced by a bounded-state training procedure. This replacement trades preparation count against memory, rather than assuming that the training computation has already been performed. The elementary block mechanism belongs to the classical finite-statistics theory of Cover [24]; the result below tracks a finite horizon, a uniform indifference band, rational precision, the whole validation phase, and a compact nonconvex candidate image.

Lemma 12.1 (A finite-horizon four-state comparison). *Let Z_i be independent Bernoulli observations of unknown mean u . For $0 < \theta < 1/2$ and $0 < \delta < 1$, put*

$$L = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad R = \left\lceil 2^L \log(2/\delta) \right\rceil. \quad (12.1)$$

There is a deterministic clocked machine with four states at every observation cut, using exactly $2LR$ observations, that decides the sign of $u - 1/2$ with error at most δ whenever $|u - 1/2| \geq \theta$. No assertion is imposed within the indifference band.

Proof. Partition the observations into R pairs of blocks, each block of length L . A positive block succeeds when all its observations are one; the following negative block succeeds when all are zero. Stop updating the decision at the first successful block and retain its sign. If none succeeds, decide negative at the end. The prescribed remaining observations are still consumed, but ignored.

Two states are absorbing positive and negative decisions. The other two mean undecided with the current block still alive or already failed. At a block boundary the clock resets the undecided flag to alive. Its polarity and position are supplied by the clock; they are not inferred from an unrecorded sample. The last observation of each block either latches the sign or starts the next block. Thus four states suffice even at the block boundaries.

Write $x = u^L$, $y = (1 - u)^L$ and $s = x + (1 - x)y$. The probability of a positive decision is exactly

$$H_{L,R}(u) = \frac{x}{s} \{1 - (1 - s)^R\}. \quad (12.2)$$

Here $s > 0$, including at $u = 0, 1$. For $u \geq 1/2 + \theta$, the error is at most $y/x + (1 - s)^R$. For $u \leq 1/2 - \theta$, it is at most x/y ; the default negative decision is then correct. Since

$$\log \frac{1/2 + \theta}{1/2 - \theta} \geq 4\theta, \quad s \geq \max(x, y) \geq 2^{-L},$$

both errors are bounded by $e^{-4\theta L} + e^{-R2^{-L}} \leq \delta$. The logarithmic inequality follows by differentiation from its equality at zero. This also proves uniformity over the two composite hypotheses. $\square$

12.1. The whole-schedule continuation value. In the one-row reset interface of Section 9, let $W_c(B, Q)$ be the least peak width over all finite training lengths and all auditors whose worst-case expected signed score is strictly greater than c . Every training-observation cut, the decision cut, every fresh random-bit cut, and both validation cuts are charged. Observations in Ω are atomic for this definition. The event is selected before the two validations, exactly as in that interface. The schedule clock and read-only constants are recorded as separate resources. The class includes randomized machines; its rational subclass uses only finite fair-bit randomizers. Write W_c^{rat} for the corresponding minimum in the latter class. Empty feasible sets have value $+\infty$.

Theorem 12.2 (Uniform end-to-end realization of a decision cover). *Let $B \subset \Delta(\Omega)$ be nonempty and compact, Q fixed, and $d = |\Omega|$. For every strict threshold c ,*

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q). \quad (12.3)$$

The last inequality is interpreted only for finite κ_c ; when $\kappa_c = +\infty$ all three quantities are infinite.

More explicitly, fix $K \geq 2$ and a response list with uniform envelope at least v . Round its coordinates to denominator 2^r , and choose $q_i \in [0, 1] \cap 2^{-r}\mathbb{Z}$ with $|q_i - Qh_i^{(r)}| \leq \rho$. For θ, δ as in Lemma 12.1, a rational auditor uses

$$N = 2LR(K - 1) \quad (12.4)$$

candidate training observations and two validation observations. Its expected score is at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)\delta. \quad (12.5)$$

Its completed-training-observation cuts have width at most $4K$, its decision cut has width K , and its entire random-bit-refined schedule has width at most $12Kd$. No empirical histogram is constructed. The response and comparison thresholds have a direct read-only representation of size $O(K^2d(r + 3) + \log L + \log R + \log K)$ bits.

Proof. Use a sequential tournament. Initially response 1 is incumbent. At stage $j = 2, \dots, K$, compare incumbent i with challenger j using fresh candidate observations. For one observation $w \sim b$, generate a Bernoulli variable with conditional success probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (12.6)$$

This lies in $[0, 1]$ and is dyadic of denominator 2^{r+2} . Its unconditional mean is $1/2 + (\tilde{f}_j(b) - \tilde{f}_i(b))/4$, where $\tilde{f}_i(b) = q_i - bh_i^{(r)}$. Conditional on the incumbent at the start of this stage, all these variables are independent and have this mean. The fixed candidate b is not changed by the comparison. Run Lemma 12.1 and replace i by j exactly when its decision is positive.

At each comparison, conditionally on all earlier stages, the probability of choosing the inferior response when their $\tilde{f}$ values differ by at least 4θ is at most δ . With probability at least $1 - (K - 1)\delta$, no such error occurs. On this event each stage loses at most 4θ relative to the better of its two responses. Induction on j gives

$$\tilde{f}_i(b) \geq \max_i \tilde{f}_i(b) - 4(K - 1)\theta.$$

Replacing the calibrated means by their true means loses at most 2ρ . Rounding the original list loses at most 2^{1-r} in its envelope, by Proposition 10.2. Any response payoff belongs to $[-1, 1]$, so failure of the good event loses at most $2(K - 1)\delta$ in expectation. This proves (12.5), uniformly over b .

During a comparison keep the incumbent index and one of the four block states. The challenger and block positions are fixed by the schedule. At the end of a comparison the sign updates the incumbent and is forgotten. This gives the $4K$ bound at completed-observation cuts. While drawing the coin in (12.6), keep this register and the current observation, and use the three-label binary sampler of Lemma 10.1. Thus at most $12Kd$ labels occur at any intervening random-bit cut. At the final decision cut only the incumbent remains. Proposition 10.2 gives at most $K(2d + 1)$ labels during event selection and $2K(d + 1)$ during validation; both are bounded by $12Kd$. The three-valued terminal output fits this bound as well. These are compatible transitions over the entire schedule.

The coin thresholds can be precomputed for all triples (i, j, w) and stored as $r + 3$ -bit integers. The event thresholds and L, R, K give the stated description bound; no arbitrary real coefficient is used by the executed machine. Formula (12.4) counts every training observation, including ignored observations after a block decision has been latched.

For (12.3), a width- W auditor has at most W labels at its decision cut. The operational upper bound of Theorem 9.2 implies $\kappa_c \leq W$. If $K = \kappa_c < \infty$, its optimal list has a positive margin over c . Take r large, $\rho \leq 2^{-r}$, and then θ, δ small enough that the loss in (12.5) is smaller than that margin. If $K = 1$, no training is needed: round the single response and use Proposition 10.2.

Rational auditors are a subclass of all auditors, proving the remaining inequality. Finally a finite-width successful machine would give finite κ_c , proving the assertion at infinity. $\square$

The dependence on the schedule is essential. The theorem is a constant-factor classification for fixed observation alphabet, not an optimal sample-complexity theorem. In particular R is exponential in L . For an autonomous implementation, a coarse bound on the number of epochs is $T \leq N(r+3) + r + 4$; Proposition 11.3 then gives at most $12Kd(T+1)$ states. The free-clock and autonomous statements are different resource bounds. Cover's theorem already exhibits the importance of this distinction [24, Theorem 2 and Section 4]; bounded-state learning itself is not a novelty claim of the present theorem.

Corollary 12.3 (Finite-precision, calibrated certificates). *Suppose the true training kernels differ from those used in an ideal auditor by at most α per training call, and the candidate and target validation laws differ by at most β_c and β_t . The right side of (12.5) is decreased by at most*

$$2 \min\{1, N\alpha\} + \beta_c + \beta_t$$

for the same implemented machine. A calibration error ρ in the known response means is already accounted for separately in that formula. When the training source is unchanged, $\alpha = 0$, regardless of how large N is.

Proof. Couple the training executions until their first mismatch, as in Theorem 11.2. The selected-event payoff has range $[-1, 1]$, giving the factor two. Conditional on a common selected event, its two validation means change by at most the two total variation errors. This uses no independence between a failed coupling and its terminal event. $\square$

13. FINITE-HORIZON SAMPLE-MEMORY BOUNDS

The minimum W_c in Theorem 12.2 permits arbitrarily long finite training. To retain the preparation constraint, let $V_{N,K}(B, Q)$ denote the supremum of the worst-case expected score over the same reset auditors, using at most N training trials and satisfying the specified register bound at each acquisition, random-bit, decision and validation cut. The observation order and the external clock are fixed parts of this interface. Write $V_{N,W}$ when every bound equals W . Padding an audit with ignored trials shows monotonicity in N . This value does not identify machines having only the same terminal dictionary.

A second realization of a response cover gives a polynomial preparation bound. Its counter is larger than the four-state block counter, and both constructions remain useful. The elementary random-walk argument below is classical; the purpose of retaining its proof is to price its use in the complete, calibrated continuation machine.

Lemma 13.1 (A truncated absorbing comparison). *Let Z_i be independent Bernoulli variables with mean u . Fix integers $m, n \geq 1$ and $0 < \theta < 1/2$. Start at $S_0 = 0$, add $2Z_i - 1$ while $|S_{i-1}| < m$, and retain the state forever after either boundary is reached. At time n declare positive exactly when $S_n \geq 0$. The machine has $2m + 1$ states. Whenever $|u - 1/2| \geq \theta$, its error probability is at most*

$$e_{m,n}(\theta) = e^{-4\theta m} + e^{-2n\theta^2}. \quad (13.1)$$

If τ is the boundary hitting time without truncation, then, with $x = 2u - 1$,

$$\mathbb{E}_u \tau = \begin{cases} m \tanh(m \operatorname{arctanh} x) / x, & 0 < |x| < 1, \\ m^2, & x = 0, \\ m, & |x| = 1. \end{cases} \quad (13.2)$$

In particular $\mathbb{E}_u(\tau \wedge n) \leq \min(n, m^2)$, and outside the indifference band it is at most $\min(n, m^2, m/(2\theta))$.

Proof. First suppose $1/2 < u < 1$ and put $r = (1 - u)/u$. For the unabsorbed walk $T_j = \sum_{i \leq j} (2Z_i - 1)$, the process r^{T_j} is a nonnegative martingale. Stopping on $\{-m, m\}$ gives

$$\mathbb{P}_u(T_\tau = -m) = \frac{r^m}{1 + r^m} \leq r^m.$$

For completeness, absorption has finite mean: in each block of $2m$ steps, a run of $2m$ positive increments has probability $u^{2m} > 0$ and forces absorption from every interior state. Thus the unabsorbed survival time is bounded by a geometric number of such blocks. Stopping first at $\tau \wedge j$, then using bounded convergence on the stopped exponential martingale, justifies the displayed identity. The estimate $\log((1 + 2\theta)/(1 - 2\theta)) \geq 4\theta$ gives $r^m \leq e^{-4\theta m}$ when $u \geq 1/2 + \theta$.

If the walk has not been absorbed, a negative final answer requires $T_n < 0$. Hoeffding's inequality [25] for variables in $[-1, 1]$ gives

$$\mathbb{P}_u(T_n \leq 0) \leq e^{-2n(u-1/2)^2} \leq e^{-2n\theta^2}.$$

Partitioning a wrong answer according to absorption proves (13.1). Reflection proves the lower-mean case; its nonabsorbed error includes $T_n = 0$ and satisfies the same bound. At $u = 0, 1$ the assertion follows directly.

The drift martingale $T_j - xj$ stopped at $\tau \wedge j$ gives $\mathbb{E}_u T_\tau = x\mathbb{E}_u \tau$, by integrability of τ and boundedness of the stopped walk. The hitting probabilities yield $\mathbb{E}_u T_\tau = m(1 - r^m)/(1 + r^m)$. Since $r = \exp(-2 \operatorname{arctanh} x)$, this is (13.2) when $x \neq 0$. For $x = 0$, stop $T_j^2 - j$ to obtain $\mathbb{E} \tau = m^2$. The endpoint formula is deterministic. Finally $\tanh(my) \leq m \tanh y$ for integer $m \geq 1$ and $y \geq 0$: induction follows from the addition formula for $\tanh$. This proves $\mathbb{E} \tau \leq m^2$; the bound $m/|x|$ follows from $|T_\tau| \leq m$. Truncation can only decrease the mean. $\square$

Theorem 13.2 (A preparation-width family of realizations). *Let $B \subset \Delta(\Omega)$ be compact, $d = |\Omega|$, and let $h_1, \dots, h_K \in [0, 1]^\Omega$, $K \geq 2$, have envelope $v = \min_{b \in B} \max_i (Q - b)h_i$. Round their coordinates to dyadic precision r , obtaining $h_i^{(r)}$, and choose known dyadic numbers $q_i \in [0, 1]$ with $|q_i - Qh_i^{(r)}| \leq \rho$. Suppose q_i also have denominator 2^r ; enlarging r accommodates any specified dyadic calibration. For $m, n \geq 1$ and $0 < \theta < 1/2$, an explicitly clocked fair-bit audit satisfies*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1), \quad (13.3)$$

$$\inf_{b \in B} \mathbb{E}_b D \geq v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)e_{m,n}(\theta). \quad (13.4)$$

At completed training-trial cuts its width is at most $K(2m + 1)$; at the decision cut it is at most K . Every random-bit cut, the selected event, and both validations are included in W . In particular one can replace $e_{m,n}(\theta)$ by δ by taking

$$m = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad n = \left\lceil \frac{\log(2/\delta)}{2\theta^2} \right\rceil. \quad (13.5)$$

Proof. For a fixed unknown b , set $g_i(b) = q_i - bh_i^{(r)}$. Compare the current incumbent i with the next index j on fresh observations. Conditional on the observation w , generate a coin with probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (13.6)$$

This is in $[0, 1]$, is dyadic with denominator 2^{r+2} , and has mean $1/2 + (g_j(b) - g_i(b))/4$. Apply Lemma 13.1, selecting j for a positive answer and i otherwise. Conditional on all earlier stages, the observations and fair bits for this comparison are fresh. The probability of selecting the worse index when $|g_j - g_i| \geq 4\theta$ is at most $e_{m,n}(\theta)$.

Outside these exceptional comparisons each tournament stage loses at most 4θ relative to the better of its two contestants. Induction shows a total loss at most $4(K - 1)\theta$ relative to $\max_i g_i$.

The union probability of exceptional comparisons is at most $(K-1)e_{m,n}(\theta)$ even though the incumbent is random. All g_i lie in $[-1, 1]$, so their range is at most two. Consequently

$$\mathbb{E}_b g_i(b) \geq \max_i g_i(b) - 4(K-1)\theta - 2(K-1)e_{m,n}(\theta).$$

The calibration changes selection and evaluation by at most 2ρ . For each coordinate rounding, $\|h_i - h_i^{(r)}\|_\infty \leq 2^{-r}$, so $|(Q-b)(h_i - h_i^{(r)})| \leq 2^{1-r}$. A nested random event with mean $h_i^{(r)}$ has exactly that rounded payoff; hence (13.4).

At a completed trial retain the incumbent and the absorbing walk state. The stage number j is supplied by the predetermined clock, not by a sample-dependent register. During coin generation additionally retain w and at most three sampler labels, by Proposition 10.2. This gives $3dK(2m+1)$ labels. The terminal event sampler, held event, and validation storage use at most $K(2d+1)$, $K(d+1)$, and $2K(d+1)$ labels, respectively, as in Theorem 12.2; all are bounded by the stated W . The stage boundary discards the walk and initializes the next one. After absorption, remaining observations and bits in that stage are ignored, but are still charged to the fixed schedule. Thus the count N includes padding. Each active coin uses $r+2$ fair bits. A coarse bound for the full number of epochs is $T \leq N(r+3) + r + 4$.

The finite transition description consists of the dyadic coin table, the boundary and horizon integers, and the nested-event thresholds. It has $O(K^2d(r+3) + \log m + \log n + \log K)$ bits under binary encoding; the constants depend on a fixed encoding convention. No unknown b occurs in that description. The autonomous conversion stores the clock and has width at most $(T+1)W$, rather than W . Substitution of (13.5) proves the last assertion. $\square$

Corollary 13.3 (Two general preparation regimes). *For a strict K -response cover and a prescribed positive score margin, there are whole-schedule rational realizations of that margin with width at most $12dK$ and exponential block preparation count, and with width at most $3dK(2m+1)$ and preparation count $n(K-1)$ as in (13.5). Both constrain every training cut. Their resource pairs are upper bounds for $\mathbb{V}_{N,W}$, not a claim of an optimal frontier for every compact image.*

Proof. Choose rounding, calibration, band and error budgets whose sum is smaller than the strict margin. Apply Theorem 12.2 or Theorem 13.2. A one-response cover needs no training and is implemented directly by the rational event sampler. $\square$

14. AN EXACT STATISTICAL FRONTIER AT ALL CUTS

The preceding realizations are uniform upper bounds. A family with an exact answer illustrates what an all-algorithm statistical lower bound requires. Unlike exact computation of a prescribed word function, the objective here is the best decision risk among all rules, including stochastic rules, under a fixed preparation schedule and full profile.

Let $\Theta = \{1, \dots, M\}$ and fix a known prior $\pi_1 \geq \dots \geq \pi_M \geq 0$, $\sum_j \pi_j = 1$. At time t , an independent indicator E_t has known mean $q_t \in [0, 1]$. The report is $Y_t = \vartheta$ if $E_t = 1$, and $Y_t = *$ otherwise. All indicators are independent of the parameter and the machine's private randomization. The unknown parameter ϑ is fixed during the n observations. The machine starts in a parameter-independent state, has at most K_t labels after report t , and finally guesses ϑ . Its initial and final randomization is part of the same finite causal machine. The objective is Bayesian probability of a correct answer under π . Parameter-dependent loss is specified here and is not identified with the one-row physical signed-score problem.

Write $P_k = \sum_{j=1}^k \pi_j$ and put

$$k_t = \min\{M, K_t, K_{t+1}, \dots, K_n\}, \quad a = \prod_{s=1}^n (1 - q_s), \quad w_t = q_t \prod_{s=t+1}^n (1 - q_s). \quad (14.1)$$

Thus $a + \sum_t w_t = 1$ and k_t is nondecreasing in t .

Theorem 14.1 (Exact all-cut revelation frontier). *Over all stochastic causal machines satisfying the specified profile, the optimal Bayesian success probability is*

$$S_{\mathbf{q}, \mathbf{K}}(\pi) = a\pi_1 + \sum_{t=1}^n w_t P_{k_t}. \quad (14.2)$$

A single deterministic machine attains this value simultaneously for all $\mathbf{q} \in [0, 1]^n$. The upper bound therefore applies to all approximate statistical decision rules, not just to machines implementing a chosen selector exactly.

Proof. Let L be the last time with $E_L = 1$, with $L = 0$ when no revelation occurs. Then $\mathbb{P}(L = 0) = a$ and $\mathbb{P}(L = t) = w_t$. These events, and every particular pattern of indicators, are independent of ϑ .

First consider any stochastic machine. Realize all of its transition and output kernels by independent random variables indexed by time, state and input, and also realize its initial randomization. This is possible on a product space independent of ϑ and the indicators. Conditional on the entire random table the machine is deterministic. This conditioning is an analysis device, not a free persistent random seed in an implementation.

Fix that table and an indicator pattern with last revelation $t > 0$. The map from ϑ to the terminal guess factors through the state at every time $s \geq t$: all reports after t are the same symbol $*$, so the remaining transitions and output are fixed maps. The set of possible terminal guesses therefore has cardinality at most $\min_{s \geq t} K_s$, and at most M guesses can be correct. Its prior success probability is bounded by the mass of the k_t largest prior atoms, namely P_{k_t} . Earlier revelations and any parameter information encoded before t do not evade this future bottleneck. On the no-revelation pattern, the output is independent of ϑ and its success probability is at most π_1 . Integrate these bounds over the random table and indicator patterns to obtain the upper bound in (14.2). In particular no extremality claim about a nonconvex minimax set has been used.

For attainment, use the labels $\{1, \dots, k_t\}$ at time t , start with label one, and use the following deterministic updates. On $*$, hold the label. On a revelation $j \leq k_t$, replace it by j ; on a revelation $j > k_t$, replace it by one. The holding update is legal because $k_{t-1} \leq k_t$. At termination output the label. Conditional on $L = t > 0$, exactly the parameters $1, \dots, k_t$ are correct; on $L = 0$, exactly parameter one is correct. This proves equality. The updates use only the prescribed profile and prior ordering, not the probabilities q_t , establishing simultaneous attainment. $\square$

Corollary 14.2 (Sharp preparation–peak tradeoff). *For the uniform prior, a constant reveal probability $0 < q < 1$, and constant width $1 \leq K \leq M$, write $a_n = (1 - q)^n$. Then*

$$S_{n, K} = \frac{a_n + K(1 - a_n)}{M}. \quad (14.3)$$

For $n \geq 1$ and target success $s \in [1/M, 1]$, the least feasible width is

$$K_{n, s} = \max \left\{ 1, \left\lceil \frac{Ms - a_n}{1 - a_n} \right\rceil \right\}, \quad (14.4)$$

provided this integer is at most M ; otherwise the target is infeasible at that preparation count. If $s > 1/M$ and $K > Ms$, the least preparation count for width K is

$$n_{K, s} = \left\lceil \frac{\log((K - 1)/(K - Ms))}{-\log(1 - q)} \right\rceil. \quad (14.5)$$

If $K \leq Ms$ and $s > 1/M$, no finite n attains the target, except that equality can be attained when revelations are certain, a case excluded by $q < 1$. Width one gives success exactly $1/M$ for all n .

Proof. Here $P_k = k/M$ and $k_t = K$. Since $\sum_t w_t = 1 - a_n$, Theorem 14.1 gives (14.3). Solving $a_n + K(1 - a_n) \geq Ms$ first for K and then for n proves the formulas. For $K > 1$, the limiting value K/M is approached strictly from below when $0 < q < 1$, which proves the boundary assertion. $\square$

Corollary 14.3 (Exact cost of a changed checkpoint). *Replacing a profile $\mathbf{K}$ by a tighter profile $\widetilde{\mathbf{K}}$ loses exactly*

$$\sum_{t=1}^n w_t (P_{k_t} - P_{\widetilde{k}_t}).$$

Inserting a charged holding cut of width H immediately after time s replaces k_t by $\min(k_t, H)$ for $t \leq s$, and leaves the later terms unchanged. In particular a single narrow early cut does not constrain information acquired after that cut.

Proof. Subtract (14.2) for the two profiles. A holding cut is a further bottleneck exactly for revelations that precede it. The same deterministic construction realizes the refined schedule. $\square$

For example, with $M = 4$, $q_1 = q_2 = q_3 = 1/2$, and $(K_1, K_2, K_3) = (4, 1, 4)$, the effective profile is $(1, 1, 4)$ and the optimal uniform-prior success is $5/8$. Four labels at the terminal cut alone would instead give $29/32$. These are risks of complete experiments, not counts of residual classes of one word function. For a nonuniform prior the same theorem weights every surviving label by its actual mass. The nested optimal label sets are what makes all cut bounds jointly attainable. For a general loss or observation channel such nesting need not hold; compatible continuation updates, not independently optimized dictionaries, remain necessary.

The exact formula is Bayesian. Its upper bound also obstructs any uniform worst-case guarantee exceeding that Bayesian value, but the attaining prior-ordered machine is not asserted to be minimax. In particular an uncharged persistent random permutation cannot be used to turn its Bayesian success into a minimax statement. The prior, observation channel and all register cuts have been declared explicitly.

15. TRANSPORT PRICED BY ACTIVE UPDATES

A changed training source need not affect a latched computation at every subsequent scheduled preparation. The relevant quantity is the defect of its induced retained-state transition, not the number of raw reports that happen to be present. This distinction gives a sharper transfer bound without assuming equality of the training laws.

Theorem 15.1 (Occupation-weighted retained-law transfer). *Consider a fixed finite execution with common initial state law μ and retained-state transition kernels P_t, Q_t on the same finite state spaces, $1 \leq t \leq T$. The kernels may depend on a fixed unknown parameter; all inequalities below hold separately for each parameter. Let*

$$d_t(x) = \|P_t(x, \cdot) - Q_t(x, \cdot)\|_{\text{TV}}, \quad \mu_{t-1} = \mu P_1 \cdots P_{t-1}.$$

For the same terminal readout kernel F ,

$$\|\mu P_1 \cdots P_T F - \mu Q_1 \cdots Q_T F\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=1}^T \int d_t d\mu_{t-1} \right\}. \quad (15.1)$$

If $d_t(x) \leq \alpha_t \mathbf{1}_{A_t}(x) + \zeta_t(x)$, the right side can be replaced by the truncated sum $\sum_t \alpha_t \mathbb{P}_P(X_{t-1} \in A_t) + \sum_t \mathbb{E}_P \zeta_t(X_{t-1})$. For a readout in $[-1, 1]$ multiply the bound by two; for an indicator no factor two is needed. The reverse bound with P, Q exchanged is also valid.

Proof. For signed measures, expand the product difference in the order

$$\mu P_1 \cdots P_T - \mu Q_1 \cdots Q_T = \sum_{t=1}^T \mu P_1 \cdots P_{t-1} (P_t - Q_t) Q_{t+1} \cdots Q_T.$$

Integration of a row difference against μ_{t-1} has total variation at most $\int d_t d\mu_{t-1}$. All suffix kernels and F contract total variation. The triangle inequality and the bound one for probability-law

distance prove (15.1). Substitute the rowwise estimate for its occupation version. The expectation bounds follow from the oscillation of the readout. $\square$

Here the retained state must support the displayed transition kernels. For independent reset observations and a time-inhomogeneous finite machine this is automatic, even when the selected row depends on the state. For a non-reset physical process one must include the necessary analysis state or use its history kernels; ignoring a report need not remove its influence on subsequent physical dynamics. No Markov closure of an arbitrary compressed physical state is assumed.

Corollary 15.2 (Calibrated training with absorption). *Run the audit of Theorem 13.2 under an ideal reset source and a perturbed reset source whose row laws differ by at most α per active call. Use the same implemented coin and event randomizers. If A is the number of active comparison updates under the ideal execution, the change in its selected-response score is at most*

$$2 \min(1, \alpha \mathbb{E}A) + \beta_c + \beta_t, \quad \mathbb{E}A \leq (K - 1) \min(n, m^2), \quad (15.2)$$

where β_c, β_t bound candidate and target validation-law changes. A comparison with conditional coin mean u has the sharper active expectation $\mathbb{E}_u(\tau \wedge n)$ in (13.2). Uniform gaps $|u - 1/2| \geq \eta$ give the bound $\min(n, m^2, m/(2\eta))$ for that comparison.

Proof. At a completed comparison trial, marginalize the fresh coin bits into its transition kernel. Before absorption, data processing bounds its row defect by the source defect α . After absorption its retained transition is the same identity under both sources; its defect is zero. At predetermined stage boundaries the common reset of the walk is also defect-free. Theorem 15.1 therefore bounds the terminal incumbent-law distance by $\alpha \mathbb{E}A$. The score associated with an incumbent lies in $[-1, 1]$. Conditional on a common incumbent and common nested event, the two validation expectations change by at most $\beta_c + \beta_t$. This proves the score estimate. Condition on the incumbent at each stage and apply Lemma 13.1; summing proves the active-count bound. Marginalizing the coin here is a proof of a law estimate, not a waiver of its implementation states, which remain charged in (13.3). $\square$

The estimate is for the output or retained computation, not for the complete raw-observation transcript. A padded sequence of N changed reports can still have total variation of order $N\alpha$ even after the machine has stopped using them. Conversely, all N preparations remain in the resource ledger even when only A updates influence the answer. Finite row-rounding defects can be inserted as the terms ζ_t in Theorem 15.1; a changed initial law adds its total variation separately. Thus source-preserving transport is the case $\alpha = 0$, while the active bound also has content for $\alpha > 0$ and does not assert generic sample-count-free stability.

16. WEIGHTED CONTINUATION OBSTRUCTIONS

We next give a distribution-sensitive version of the wordwise obstruction. It applies to finite-valued terminal decisions and is useful even when a word of small probability should not be assigned the same weight as a likely one.

Let $f : \Sigma_1 \times \cdots \times \Sigma_L \rightarrow \mathcal{A}$ have finite range. The residual of a prefix x at cut t is the function $v \mapsto f(xv)$ on the suffix alphabet. Write $r_t(f)$ for the number of distinct residuals. A stochastic machine has a charged register at every cut; all persistent randomness is included in this register. Its wordwise error is $e(w) = \Pr(\hat{f}(w) \neq f(w))$.

Theorem 16.1 (Finite-valued realization and weighted cut bound). *Exact stochastic computation of f requires $K_t \geq r_t(f)$ at every cut. The whole residual profile is simultaneously attained by a deterministic machine.*

Fix a cut, prefixes $x_1, \dots, x_r$, and a graph on $[r]$. For each edge ij choose a suffix v_{ij} such that $f(x_i v_{ij}) \neq f(x_j v_{ij})$. Let $\mu(w) \geq 0$ be arbitrary word weights. Choose edge weights $\lambda_{ij} \geq 0$ satisfying the word capacity constraints

$$\sum_{ij: w=x_i v_{ij} \text{ or } w=x_j v_{ij}} \lambda_{ij} \leq \mu(w) \quad \text{for every word } w. \quad (16.1)$$

Every machine with K states at this cut satisfies

$$\sum_w \mu(w) e(w) \geq \min_{\phi: [r] \rightarrow [K]} \sum_{ij: \phi(i)=\phi(j)} \lambda_{ij}. \quad (16.2)$$

The right side is a weighted monochromatic-edge obstruction. Maximizing it over admissible witnesses and weights gives a finite, distribution-sensitive lower bound. The bound can be an equality with positive error.

Proof. Let $a_i(m)$ be the state distribution after prefix x_i . For a fixed edge and a common state m , the future output distribution on suffix v_{ij} is the same for both prefixes. Since their correct outputs are different, the sum of their two conditional error probabilities is at least one. Multiplying by the smaller state mass and summing gives

$$e(x_i v_{ij}) + e(x_j v_{ij}) \geq \sum_{m=1}^K \min\{a_i(m), a_j(m)\}. \quad (16.3)$$

Weight the inequalities by λ_{ij} and use (16.1). The resulting lower bound is

$$F(a_1, \dots, a_r) = \sum_{ij} \lambda_{ij} \sum_m \min\{a_i(m), a_j(m)\}.$$

This is a concave continuous function on $(\Delta_K)^r$. Every point of that product is a convex combination of its deterministic vertices; concavity implies that its value is at least the minimum over those vertices. A vertex is a labeling $\phi: [r] \rightarrow [K]$, and its value is exactly the right side of (16.2). This proves the bound without treating stochastic encodings as deterministic encodings.

For zero error, choose one distinguishing suffix for every two different residuals. Equation (16.3) forces the corresponding state supports to be disjoint. Hence at least $r_t(f)$ states are needed. For the converse, keep the residual class of the observed prefix. Equal residuals have equal shifts after every next input, so the next class is well defined. At the last epoch the residual is the output label itself. These quotient transitions realize all widths simultaneously.

For a positive-error equality, let the only relevant words be two one-letter prefixes with different required labels, weighted p and $1 - p$, where $0 < p < 1$. Put $K = 1$ and $\lambda_{12} = \min(p, 1 - p)$. The cut obstruction is $\min(p, 1 - p)$, attained by always emitting the more likely label. Thus the weighted constant is sometimes exact; universal sharpness for all graphs or input distributions is not asserted. With unit weights on the same two words the total-error lower bound is one, also attained by a constant output; this is equality in the earlier $(r - K)/(r - 1)$ bound at $r = 2$, $K = 1$. $\square$

Corollary 16.2 (A Bayes-regret obstruction for constrained audits). *Fix a finite action set $\mathcal{A}$, candidate class B , and a probability measure ν on B . For a training word w , let $p_w(b)$ be its likelihood and let $f_a(b)$ be the expected validation payoff. Put*

$$R_w(a) = \int_B p_w(b) f_a(b) d\nu(b), \quad S_\nu = \sum_w \max_a R_w(a).$$

Choose a maximizing label $f(w)$. Set $\mu(w) = R_w(f(w)) - \max_{a \neq f(w)} R_w(a)$, taking zero when a maximum is tied. Let $\Gamma_{t,K}(f, \mu)$ be any lower bound supplied by (16.2). Then every width- K audit

at cut t has worst-case expected payoff at most

$$S_\nu - \Gamma_{t,K}(f, \mu). \quad (16.4)$$

Proof. The conditional distribution of the auditor's output for a fixed word is independent of the unknown candidate. Its Bayes payoff is therefore $\sum_{w,a} R_w(a) \Pr(\hat{f}(w) = a)$. Subtracting this from S_ν gives at least $\sum_w \mu(w)e(w)$. Apply the theorem, and use that a minimum over candidates is at most the average under ν . $\square$

The corollary does not exchange a maximum and minimum over a nonconvex fixed-width machine class. The prior is a lower-bound device, not a randomization resource supplied to the candidate. With a unique maximizing label and strictly positive margins at every word, equality with S_ν forces exact wordwise computation. This makes explicit the uniqueness step used in the odd-sample binary theorem below. For even sample size, the balanced-count words have zero likelihood gap. A rule can differ on those ties without losing value; equality of values then does not force one prescribed total Boolean rule, and the odd-sample width profile cannot simply be asserted for that case.

17. JOINT RESIDUAL BUDGETS AND GROWING FAMILIES

We now put lower bounds on several cuts into one statistical budget. Summing separate cut bounds without controlling their common words would count the same regret more than once.

Theorem 17.1 (A shared-capacity multicut obstruction). *Let $f : \prod_{i=1}^L \Sigma_i \rightarrow \mathcal{A}$ and $\mu(w) \geq 0$ be as in Theorem 16.1. At each chosen cut t , choose a finite prefix graph with an edge $e = (u, v)$ carrying a distinguishing suffix s_e , and a weight $\lambda_{t,e} \geq 0$. Impose the single family of constraints*

$$\sum_{t,e: w=us_e \text{ or } w=vs_e} \lambda_{t,e} \leq \mu(w) \quad \text{for every complete word } w. \quad (17.1)$$

For a graph at cut t , write

$$\Gamma_{t,K_t}(\lambda) = \min_{\phi: \text{prefixes} \rightarrow [K_t]} \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Every stochastic machine of profile $\mathbf{K}$ satisfies

$$\sum_w \mu(w) \Pr(\hat{f}(w) \neq f(w)) \geq \sum_t \Gamma_{t,K_t}(\lambda). \quad (17.2)$$

Consequently, with R_w, S_ν, μ defined in Corollary 16.2, every audit of this profile and the fixed training schedule has worst-case payoff at most

$$S_\nu - \sup_{\lambda \text{ satisfying (17.1)}} \sum_t \Gamma_{t,K_t}(\lambda). \quad (17.3)$$

For fixed graphs the supremum is the value of a finite linear program.

Proof. For each edge use the overlap inequality (16.3). Multiply it by $\lambda_{t,e}$ and sum over all cuts and edges. The left side is bounded above by the left side of (17.2), by the shared capacity constraint. At each cut the resulting overlap expression is concave in its prefix state distributions, hence at least its minimum over deterministic labelings, as proved in Theorem 16.1. Adding these bounds proves (17.2). Although prefix distributions at different cuts are compatible, minimizing them separately only weakens the bound; it does not assume independent realizability. The Bayes-regret argument of Corollary 16.2 proves (17.3).

For the last assertion introduce variables z_t and impose, for each labeling ϕ of each finite graph,

$$z_t \leq \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Maximize $\sum_t z_t$ subject to these inequalities, nonnegative edge weights, and (17.1). There are finitely many constraints. This is a computable obstruction, not a claim that every multicut statistical optimum equals this relaxation. $\square$

17.1. Actual physical word weights. For n all-on trials of the private marked collision subfamily, a complete training word w specifies the two report bits and the mark of each trial. With N_{ab} its report-pair counts,

$$p_w(p, q) = 2^{-n} p^{N_{10}+N_{11}} (1-p)^{N_{00}+N_{01}} q^{N_{01}+N_{11}} (1-q)^{N_{00}+N_{10}}. \quad (17.4)$$

The formula has the continuous endpoint interpretation $0^0 = 1$. For any finitely supported prior on actual pairs $(p, q) \in [0, 1]^2$, use this likelihood in $R_w(A) = \int p_w(p, q)(Q - b(p, q))\mathbf{1}_A d\nu$, with all marked events $A \subseteq \Omega$ as actions. Randomized events are included by linearity. Equations (17.1) and (17.3) then give a finite all-algorithm obstruction for that full training profile, with the true physical word probabilities. Changing only Q by total variation β changes every audit's payoff by at most β .

This specialization bounds all stochastic selectors on the fixed all-on training schedule, not merely exact implementations of a majority rule. It does not assert a new numerical lower bound above three for every adaptive physical audit. Adaptive training has controller-dependent word likelihoods and must instead be included in the full experiment of (11.3); using (17.4) unchanged for such a likelihood would be incorrect. The feedback-uniform lower bound used below remains Lemma 29.2.

17.2. A family with matching end-to-end growth.

Theorem 17.2 (Point-source decision spectra). *Let $d \geq 2$, $B_d = \{\delta_1, \dots, \delta_d\}$ on $[d]$, and let Q_d be uniform. For every $K \geq 1$,*

$$\mathfrak{V}_K(B_d, Q_d) = 1 - \frac{\lceil d/K \rceil}{d}. \quad (17.5)$$

For $1 - 2/d < c < 1 - 1/d$, the decision complexity is exactly d , and the whole-schedule peak satisfies

$$d \leq W_c(B_d, Q_d) \leq W_c^{\text{rat}}(B_d, Q_d) \leq 2d.$$

The upper bound uses one training observation and two validations, with no randomization. Thus the end-to-end width grows linearly in this nontrivial sequence of compact behavior images.

Proof. Assign each point source to a response that maximizes its score. Some response is assigned a set S of at least $\lceil d/K \rceil$ indices. For $s = |S| < d$, the mean of its scores on S equals

$$\frac{d-s}{d} \left(\frac{1}{d-s} \sum_{i \notin S} h_i - \frac{1}{s} \sum_{i \in S} h_i \right) \leq 1 - \frac{s}{d}.$$

For $s = d$ the mean is zero. At least one assigned source has score no larger than this mean, proving the upper bound. Partition $[d]$ into at most K groups of size at most $\lceil d/K \rceil$ and use their complements as events. This attains the lower bound.

In the stated interval the formula forces $K \geq d$. A single training observation identifies the point source i exactly. Keep i and choose the complement of $\{i\}$. During validation retain i and the candidate indicator, at most $2d$ states, then emit the target indicator minus it. This computes the prescribed signed score on every input word, including words that have zero probability under the point-source family. Its expected score is $1 - 1/d$. The lower bound follows from the necessary decision-cut width. $\square$

17.3. A sharp simultaneous storage profile. A second class separates overlapping retention obligations. Independent uniform labels $X_j \in [q_j]$ are revealed at scheduled epochs a_j and must be reported at epochs $b_j > a_j$. No later input reveals an active label, and previous outputs cannot be read as external memory. Let $A_t = \{j : a_j \leq t < b_j\}$ and $D_t = \prod_{j \in A_t} q_j$.

Theorem 17.3 (Overlapping retention intervals). *Every stochastic machine whose probability of correctly reporting all labels is at least $1 - \varepsilon$ has*

$$K_t \geq (1 - \varepsilon)D_t \quad \text{at every cut } t. \quad (17.6)$$

The exact profile $K_t = D_t$ is attained simultaneously by a deterministic machine. Thus the minimum peak number of bits lies between $\log_2 \max_t D_t + \log_2(1 - \varepsilon)$ and $\lceil \log_2 \max_t D_t \rceil$ for $\varepsilon < 1$.

Proof. Fix a cut and give the decoder, as additional information, all labels outside A_t , whether past or future. They are independent of the uniform active vector. Conditional on them, every future prediction of that vector factors through the K_t -valued retained state and fresh randomness. If $a(m | x)$ is its conditional encoder and $d(x | m)$ the induced joint prediction law, its success probability is at most

$$\frac{1}{D_t} \sum_m \sum_x a(m | x) d(x | m) \leq \frac{1}{D_t} \sum_m \sum_x d(x | m) = \frac{K_t}{D_t}.$$

Correctness of all reports implies correctness of the active reports, so averaging over the independent additional information proves (17.6). For the upper bound, keep precisely the vector of active labels, append a label when revealed, and delete it when reported. Every update is deterministic and respects the schedule. The bit bounds follow by taking logarithms and rounding for an actual binary register. The lower bound is a counting argument, not an independence assumption about labels after conditioning on a compressed state. $\square$

18. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

18.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (18.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 18.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (18.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (18.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (18.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given, this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1) \sum_j d_j,$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (18.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E} (\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E} \rho \leq (\sum_j \mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (18.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a+Jn+1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (18.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 18.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 18.1.*

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

18.2. A matching uniform sampling obstruction.

Theorem 18.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (18.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (18.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (18.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (18.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

18.3. Transport of one fixed audit.

Proposition 18.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (18.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (18.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

19. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 19.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (19.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (19.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (19.2) prove (19.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 8.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

19.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 18.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 19.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (19.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (19.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 18.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (19.3). The exact realization in Theorem 19.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (19.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (19.1) into (19.5) gives (19.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (19.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 18.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

20. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 21.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 7.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector

fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 20.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (20.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (20.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy-Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (20.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (20.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 20.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk-Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 20.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (20.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (20.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 20.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (20.2) whose identity is checked by (20.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 20.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 20.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 21.3; its verification is a different cost.

20.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 20.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 20.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

21. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

21.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 7, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 21.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{21.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

21.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 21.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (21.2)$$

For $n \geq 0$ let $\beta_{\alpha,n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (21.3)$$

Theorem 21.3 (Intrinsic adaptive degree and support bounds). *The maxima in (21.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (21.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (21.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (21.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (21.5) cannot be replaced by its behavior barycenter. For $n = 0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 20.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 21.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

21.3. A three-weight certificate invisible to constant tests.

Proposition 21.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (21.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (21.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 18 imposes an explicit finite-data restriction.

21.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 21.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

22. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 21.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 22.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine*

dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (22.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (22.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

23. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

23.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (23.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 23.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^P(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (23.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4, a$ or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the

private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

23.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (23.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), \quad a_0 \in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (23.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 23.2 (Uniform transverse collision signal). *For every preparation in (23.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (23.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (23.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma\xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A\mathbf{1}_{\{h(v_{1,y}(3)) + \sigma\xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 23.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (23.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 23.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

23.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 23.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 21.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (23.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

24. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 24.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (24.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (24.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (24.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (24.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (24.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 24.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 24.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

24.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 24.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (24.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (24.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (24.3). Applying Theorem 24.1 proves (24.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (24.3). That is the resource-comparison consequence. The latent-preserving coupling in (24.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time

and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

25. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

25.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (25.1)$$

$$dP_n = L_T^n dR. \quad (25.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 25.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (25.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (25.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) \mid \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (25.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 24.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (25.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (25.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1}\{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 | z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 | z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (25.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f | \mathcal{F}_t]$ proves (25.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 24.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (25.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 24.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

25.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 25.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (25.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (25.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (25.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) | \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (25.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

25.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n,e^{\gamma f}}}{\ell_t^n}. \quad (25.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 25.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (25.10)$$

On the coupling of Theorem 25.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (25.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n,e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (25.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n/\ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (25.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n/N_t^n - b_t^n/\ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma|Z_t^n|^2/2$, which proves (25.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (25.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 24.1 to $g_0 = (e^{\gamma f} - e^{-\gamma})/(e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (25.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (25.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r|a^n - a| + (M/\sigma)r^3|N^n - N|.$$

The supremum estimates (25.5), followed by integration in time, prove (25.12). The same inequality applies to b^n/ℓ^n .

Finally $e^{-\gamma\ell^n} \leq N^n \leq e^{\gamma\ell^n}$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 25.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 24.1, Cauchy-Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

25.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 7.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 18.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 25.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 25.1 remove the need for that inference in the observation class treated here.

26. OBSERVATION DENSITIES AND COUPLED PROCESS VERSIONS

We record three measure-theoretic facts used in the global transport argument. Their purpose is to make explicit the passage from a tilted physical law to an observation law and from two marginal processes to one joint realization.

Lemma 26.1 (Observation-level return from a bounded tilt). *Let $dP^g/dP = g(Z)/c$, where $c = \mathbb{E}_P g$ and $e^{-\gamma} \leq g \leq e^\gamma$. For any observation sub- σ -field $\mathcal{F}$,*

$$\frac{d(P|_{\mathcal{F}})}{d(P^g|_{\mathcal{F}})} = \mathbb{E}_{P^g} \left[\frac{c}{g} \middle| \mathcal{F} \right] = \frac{c}{\mathbb{E}_P[g | \mathcal{F}]} \quad (26.1)$$

Both this density and its reciprocal lie in $[e^{-2\gamma}, e^{2\gamma}]$. In particular, for any nonnegative $\mathcal{F}$ -measurable V , $\mathbb{E}_P V \leq e^{2\gamma} \mathbb{E}_{P^g} V$.

Proof. For $A \in \mathcal{F}$, integrate c/g over A under P^g to obtain $P(A)$. Conditioning proves the first identity. In the other direction, $d(P^g|_{\mathcal{F}})/d(P|_{\mathcal{F}}) = \mathbb{E}_P[g | \mathcal{F}]/c$; equivalence and inversion prove the second. The bounds follow from the bounds on g , its conditional mean, and c . $\square$

Lemma 26.2 (Domination of the common-path measure). *Let (Y', Y) be a coupling on a standard Borel path space, with marginal laws P', P , and let $E = \{Y' = Y\}$. The subprobability measure $\kappa(A) = \Pr(E, Y \in A)$ is dominated by both P and P' . Consequently, for any nonnegative measurable path functional V ,*

$$\mathbb{E}[\mathbf{1}_E V(Y)] \leq \mathbb{E}_P V, \quad \mathbb{E}[\mathbf{1}_E V(Y')] \leq \mathbb{E}_{P'} V.$$

The assertion also holds on a common latent–observation space when the latent coordinate is shared by the coupling.

Proof. The event defining $\kappa(A)$ is contained in $\{Y \in A\}$ and, on E , in $\{Y' \in A\}$. This proves domination on sets. Integration of simple functions followed by monotone convergence proves the rest. Apply the same argument to the product coordinate for the last assertion. $\square$

Lemma 26.3 (Marginal stochastic integrals on an arbitrary coupling). *On a canonical standard Borel observation path space, suppose that under each of two marginal laws there are a continuous adapted process U , a predictable square-integrable process Z , and an innovation I such that, in that marginal's own filtration,*

$$\int_0^t Z_s dI_s = U_t - U_0 + \frac{\gamma}{2} \int_0^t |Z_s|^2 ds, \quad 0 \leq t \leq T. \quad (26.2)$$

Assume Z is bounded, as in the entropic model here. There are measurable versions of the two continuous integral processes which can be pulled back to any coupling of the observation laws. Each pulled-back version has the correct marginal law and satisfies its own identity (26.2). Neither integral thereby acquires a martingale or Itô-integral interpretation in the enlarged joint filtration.

Proof. A continuous process is determined by its values at rational times. Completed-measurable random variables on the canonical space have Borel versions. Choose such versions at the countably many rational times and extend by continuity on the measurable set where a continuous extension exists, using the zero path elsewhere. This gives a Borel $C([0, T])$ -valued version of U . Likewise take a jointly measurable version of Z modulo $dt \otimes P$; its bounded time integral has a continuous path, and its rational-time values give a measurable continuous version by the same construction. The right side of (26.2) is consequently a measurable continuous path functional. Marginal equality at all rational times, and continuity of the stochastic integral, make it indistinguishable from that integral. Repeat for the other marginal. Pullback along the two coordinate maps of any coupling preserves their respective almost-sure identities and laws. No enlargement theorem for innovations is used. $\square$

In Theorem 27.2, Lemma 26.1 is applied with $\mathcal{F} = \mathcal{F}_T$ to the integrated squared coefficient comparison. Lemma 26.2 justifies restricting a same-path estimate to the equality event of the latent-preserving coupling. Lemma 26.3 fixes the meaning of the coupled backward martingale integrals before their difference is estimated. Thus the coupled notation retains the marginal observation filtrations throughout.

27. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 25. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 27.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S |\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]| \leq 4M\epsilon, \quad (27.1)$$

$$\mathbb{E}_S |\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]|^2 \leq 8M^2\epsilon. \quad (27.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u \mathbf{1}_A], \mathbb{E}_Q[u \mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by more than M . Integrating the displayed identity proves (27.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 25.2 and Theorem 25.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2} \{2\eta_n^2 + 16M^2T\epsilon_n\}, \quad (27.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (27.4)$$

where $\epsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 27.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 25.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (27.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 25.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2 T \varepsilon_n / \sigma^2, \quad (27.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2 T \varepsilon_n / (\gamma^2 \sigma^2). \quad (27.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2 \mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2 T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (27.8)$$

The first term on the right has the explicit bound in (25.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) | \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) | \mathcal{F}_t].$$

Lemma 27.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2 \varepsilon_n + 2\mathbb{E}_\nu |H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (27.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma} \varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma} \eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 25.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) | \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma} A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma} A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (27.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (27.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the

corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (27.9) and integrate for time T . This proves (27.6)–(27.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (27.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (27.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 25.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

28. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 23.1 and Lemma 23.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (23.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (28.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 28.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (28.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 23.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (28.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 28.2 (Microscopic-to-audit and conditional-law composition). *For the single family (28.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \quad (28.3)$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \quad (28.4)$$

The joint preparation–continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 25.2, Theorem 25.3, and Corollary 25.4.

(d) On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 22.1 and 21.6.

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (28.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 23.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 23.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 18.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (28.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (28.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 28.1 gives the signal-distance hypothesis of Theorem 25.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (28.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous

observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 18.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 23.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 28.2 and the preceding results prove Theorem C.3. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

29. SHARP DECISION MEMORY FOR THE COLLISION EXPERIMENT

We now apply the continuation spectrum to the same marked experiment that supplies the microscopic certificate. Both its impossibility statement and its construction concern the same score, the same private candidate family, and the same training-validation checkpoint.

29.1. The marked family and its first two spectral values. In the all-on row of Section 28, a private width-one candidate has two independent report bits with probabilities $p, q \in [0, 1]$, independently of the fair actual mark. Denote its law by $b(p, q)$ and put

$$a = (1 - p)(1 - q), \quad c = pq.$$

Let $B = \{b(p, q) : p, q \in [0, 1]\}$. The ideal target Q_* is supported on 000, 001, 110, 111, with masses $5/16, 3/16, 3/16, 5/16$, respectively. On these four atoms the candidate masses are $a/2, a/2, c/2, c/2$. The elementary continuation constraint implies

$$\sqrt{a} + \sqrt{c} \leq 1, \quad \min(a, c) \leq \frac{1}{4}. \quad (29.1)$$

Indeed the first inequality is Cauchy-Schwarz applied to $(\sqrt{1-p}, \sqrt{p})$ and $(\sqrt{1-q}, \sqrt{q})$; the second follows from the first. If $a \geq c$, then $c \leq 1/4$ and

$$a + 3c \leq (1 - \sqrt{c})^2 + 3c = 1 - 2\sqrt{c} + 4c \leq 1. \quad (29.2)$$

The symmetric inequality holds when $c \geq a$.

Theorem 29.1 (A sharp two-state obstruction). *For the ideal marked collision family,*

$$\mathfrak{V}_1(B, Q_*) = \frac{1}{8}, \quad \mathfrak{V}_2(B, Q_*) = \frac{3}{8}. \quad (29.3)$$

If $\|Q - Q_\|_{\text{TV}} \leq \beta$, every reset auditor retaining at most two states at the training-validation cut has worst-case expected score at most $3/8 + \beta$, regardless of its finite training length and its upstream or downstream memory. The assertion allows mark-dependent validation events and stochastic state transitions.*

Proof. For one state, use the event $\{000, 111\}$. Its score is $5/8 - (a + c)/2 \geq 1/8$. Conversely, average the two candidates with $(a, c) = (1, 0)$ and $(0, 1)$. Their average is uniform on the four supported atoms and is at total variation distance $1/8$ from Q_* . Every fixed response has average score at most $1/8$, so its minimum over the two candidates is at most $1/8$. This proves the first equality; the average is a bound on scores, not an allowed private candidate.

For the upper bound with two states, consider the three candidates $b_0 = b(0, 0)$, $b_1 = b(1, 1)$ and $b_* = b(1/2, 1/2)$. A response may be set to zero outside the support of Q_* without decreasing its payoff at any candidate. Write its four remaining coordinates as (x, y, z, t) . Its three payoffs are

$$\begin{aligned} f_0 &= (-3x - 5y + 3z + 5t)/16, \\ f_1 &= (5x + 3y - 5z - 3t)/16, \\ f_* &= (3x + y + z + 3t)/16. \end{aligned}$$

Since all four coordinates are in $[0, 1]$,

$$f_0 + f_1 \leq \frac{1}{4}, \quad f_0 + f_* \leq \frac{3}{4}, \quad f_1 + f_* \leq \frac{3}{4}. \quad (29.4)$$

Suppose two responses covered all three candidates with payoff strictly greater than $3/8$. The first inequality forces different responses to cover b_0 and b_1 . The other two inequalities then force both responses to have payoff less than $3/8$ at b_* . This is a contradiction. Hence $\mathfrak{V}_2 \leq 3/8$.

For attainment take the responses, in the same four-atom order,

$$h_0 = (2/3, 0, 1, 1), \quad h_1 = (1, 1, 0, 2/3),$$

with zero response elsewhere. Both have target mean $17/24$. Their payoffs are $17/24 - a/3 - c$ and $17/24 - a - c/3$. If $a \geq c$, the first is larger and (29.2) makes it at least $17/24 - 1/3 = 3/8$. The other case is symmetric. This proves equality. For the perturbed target, apply (9.4) and the operational upper bound of Theorem 9.2. $\square$

The obstruction has an immediate continuation interpretation. Each post-training state supplies one validation response. The three witness candidates in the proof cannot be covered above $3/8$ by two such responses. No increase of sample size can compensate for losing the third continuation at that cut. This argument does not use the unrelated Bernoulli majority calibration.

Lemma 29.2 (Feedback completion of the physical lower bound). *The upper bound $3/8 + \beta$ in Theorem 29.1 remains valid when the auditor may train on any of the original feedback rows and may select a row-event pair after the decision cut. Here every target row is the declared feedback pushforward of Q , with $\|Q - Q_*\|_{\text{TV}} \leq \beta$. In particular, two states cannot evade the bound by using a different sensor gate during training or validation.*

Proof. It suffices to restrict the adversary to an admissible subfamily. For $p, q \in [0, 1]$, let its first emission be $X_1 \sim \text{Ber}(p)$, and, after gate A , let its second emission be AX_2 , with an independent $X_2 \sim \text{Ber}(q)$ and the independent fair actual mark. These are genuine width-one private rows: the gate is a current input, and the second emission uses no retained copy of X_1 .

For any deterministic gate policy $g : \{0, 1\} \rightarrow \{0, 1\}$, the marked row is the pushforward of $b(p, q)$ under $T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$. The physical target row is the pushforward of Q by the same map, since acquisition does not act on the path. Including the gate in the transcript

changes nothing: it is also a deterministic function of the displayed coordinates. For a selected row event, the response pulls back to $\mathbf{1}_A \circ T_g$ on the all-on alphabet. A randomized row-event continuation pulls back to a function $h_m \in [0, 1]^\Omega$. Hence, conditional on cut state m , its expected score on this subfamily is exactly $(Q - b(p, q))h_m$.

Whatever training rows were used, their effect is solely the distribution of the retained state. The two-state score is thus a convex combination of two such responses and is bounded above by their envelope. Apply the three-candidate argument of Theorem 29.1, and then the uniform target perturbation bound. Since this is a subfamily of the full private candidate class, it is a valid lower-bound witness for that full class. Randomized feedback is covered by conditioning on its independent random table and then averaging the pulled-back response. $\square$

29.2. A three-response construction. Define three deterministic events, all ignoring the mark,

$$A_D = \{00, 11\} \times \{0, 1\}, \quad A_0 = \{00\} \times \{0, 1\}, \quad A_1 = \{11\} \times \{0, 1\}.$$

Their ideal scores are $1 - a - c$, $1/2 - a$ and $1/2 - c$.

Lemma 29.3 (The three-response envelope). *For the original private candidate image,*

$$\min_{p, q} \max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = \sqrt{2} - 1. \quad (29.5)$$

In particular $\mathfrak{V}_3(B, Q_) \geq \sqrt{2} - 1$.*

Proof. By symmetry suppose $a \geq c$. Then $c \leq 1/4$ and the maximum is $1 - a - c$ for $a \leq 1/2$, and $1/2 - c$ for $a \geq 1/2$. If $a \leq 1/4$, the first expression is at least $1/2$. If $1/4 \leq a \leq 1/2$, (29.1) gives

$$a + c \leq 1 - 2\sqrt{a} + 2a \leq 2 - \sqrt{2},$$

since the last function is increasing on $[1/4, 1/2]$. If $a \geq 1/2$, then $c \leq (1 - \sqrt{a})^2 \leq (1 - 1/\sqrt{2})^2 = 3/2 - \sqrt{2}$. Both cases give the asserted lower bound. It is attained at $p = q = 1 - 1/\sqrt{2}$, where $a = 1/2$ and $c = 3/2 - \sqrt{2}$. The symmetric point gives the same value. $\square$

Lemma 29.4 (Odd-sample majority selection with a uniform regret bound). *Let $n = 2h - 1$ be odd. In n independent candidate trials, let C_0, C_1 count report pairs 00, 11. Choose A_1 if $C_0 \geq h$, choose A_0 if $C_1 \geq h$, and otherwise choose A_D . The first two cases are mutually exclusive. Its expected ideal score satisfies*

$$g_n(p, q) \geq \sqrt{2} - 1 - \frac{1}{4\sqrt{en}} - \frac{1}{4}e^{-n/8}. \quad (29.6)$$

Its event label has three possible values at the decision cut.

Proof. Because $a + c \leq 1$,

$$\max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = 1 - a - c + (a - 1/2)_+ + (c - 1/2)_+.$$

The selected payoff is $1 - a - c + (a - 1/2)\mathbf{1}_{\{C_0 \geq h\}} + (c - 1/2)\mathbf{1}_{\{C_1 \geq h\}}$. Its expected loss from this envelope is exactly the sum, for $u = a, c$, of $|u - 1/2|$ times the probability that the binomial majority selects the wrong side of $u = 1/2$. At $u = 1/2$ the contribution is zero. No independence between C_0 and C_1 is used.

For $0 < u < 1/2$ and $k \geq h$, the ratio of the binomial mass at u to that at $1/2$ is

$$[4u(1 - u)]^{n/2} \left(\frac{u}{1 - u} \right)^{k - n/2} \leq [4u(1 - u)]^{n/2}.$$

The fair binomial upper tail is exactly $1/2$ because n is odd. Reflecting k treats $u > 1/2$. The endpoint cases follow directly or by continuity. Thus the wrong-side probability is at most

$$\tfrac{1}{2}[1 - 4(u - 1/2)^2]^{n/2} \leq \tfrac{1}{2}e^{-2n(u - 1/2)^2}.$$

For $x \geq 0$, the maximum of $xe^{-2nx^2}/2$ is $1/(4\sqrt{en})$, reached at $x = 1/(2\sqrt{n})$. At least one of a, c is at most $1/4$ by (29.1); that coordinate has $|u - 1/2| \in [1/4, 1/2]$ and contributes at most $e^{-n/8}/4$. Bound the other coordinate by the unrestricted maximum and use Lemma 29.3. $\square$

29.3. Simultaneous exact widths and the full acquisition schedule. The preceding construction admits more compression than storing its two raw counts. Write 0, 1, * for the three report-pair categories 00, 11 and the two off-diagonal pairs. For $n = 2h - 1$, let F_n be the terminal event-label function of this word. At a trial cut t , let $r = n - t$ and let (u, v) be the two prefix counts. Associate the residual key

$$\mathcal{R}_t(u, v) = \begin{cases} \text{L}, & u \geq h, \\ \text{R}, & v \geq h, \\ (\min(h - u, r + 1), \min(h - v, r + 1)), & u, v < h. \end{cases} \quad (29.7)$$

The symbols L, R mean the constant decisions A_1, A_0 . In a pair key, a coordinate $r + 1$ means that this majority is no longer reachable. The pair $(r + 1, r + 1)$ represents the constant decision A_D .

Theorem 29.5 (Exact profile of the physical audit rule). *The keys in (29.7) are exactly the distinct continuation residuals of F_n . Their number at trial cut t is*

$$r_t(F_n) = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq h - 1, \\ \binom{n-t+3}{2}, & h \leq t \leq n. \end{cases} \quad (29.8)$$

This is the minimum simultaneous width profile for exact computation of this rule, for deterministic and stochastic machines alike. Its maximum is $\binom{h+2}{2}$ at $t = h$.

For $n = 399$, the maximum trial-cut width is 20301, and the terminal training cut has exactly three states. Acquiring each report pair one bit at a time, discarding its training mark, and then executing independent candidate and target validations gives a deterministic machine with at most $40602 < 2^{16}$ states at every scheduled epoch. One additional ignored training trial may be inserted to give the same 400 training slots and 402 total trials as the earlier certificate. It changes neither the payoff nor these width bounds.

Proof. If a majority has already been reached, its label cannot change. Otherwise, the first coordinate of the pair key is exactly the number of additional 0 symbols needed, with the unreachable threshold clipped to $r + 1$; the second coordinate has the analogous meaning. These data determine the suffix function. Two different reachable pair keys can be distinguished by a suffix containing only one of the two symbols and enough * symbols to fill the remaining positions: use a count between the two different thresholds. The other majority cannot be reached by this suffix. The constant-majority keys are distinct from every pair key because an all-* suffix returns A_D for the latter. Thus the keys are exactly the residuals.

For $t < h$, no majority has been reached and every threshold is at most $r + 1$. The distinct pairs (u, v) with $u + v \leq t$ give $\binom{t+2}{2}$ keys. For $t \geq h$, the two constant-majority keys are reachable. The remaining keys are precisely

$$(i, j) \in \{1, \dots, r + 1\}^2, \quad i + j \geq r + 1.$$

To see surjectivity, use $u = h - i, v = h - j$; these counts are nonnegative, less than h , and sum to at most t because $2h - t = r + 1$. To see necessity before clipping, use the same inequality on $h - u, h - v$; if either coordinate is clipped, the inequality remains automatic. There are $(r + 1)^2 - (r - 1)r/2$ such pairs, with the expression valid also for $r = 0, 1$. Adding the two constant keys gives $(r + 2)(r + 3)/2$. This proves (29.8). The finite-valued realization theorem, Theorem 16.1, proves both the simultaneous realization and stochastic minimality for the specified rule. The two polynomial pieces of the profile give its maximum as stated.

For an explicit transition definition, order the reachable keys lexicographically, putting L, R first. For each key choose the lexicographically first prefix-count pair representing it. After the next category, increment the appropriate count and apply $\mathcal{R}_{t+1}$. Equality of residual functions implies equality of the shifted residuals, so this definition is independent of the chosen representative. The representative is part of the time-indexed transition table, not another stored register. The clock supplies t but no sample history. All these maps are deterministic and rational.

At a bit-level epoch, the state consists of a residual key and, only between the two report bits, the first bit. It therefore has at most $2 \max_t r_t$ values. On reading the second bit, determine the category, update the key and discard both bits. The mark is read and discarded; all-on controls and resets are prescribed by the free schedule. After training, keep just one of the three event labels. During validation, keep that label, the completed candidate-event indicator, and a first-bit buffer when needed. At most $3 \cdot 2 \cdot 2 = 12$ states suffice there; a completed target pair emits its event indicator minus the stored candidate indicator. The final score has three possible values. These phase-specific alphabets may be embedded in a single alphabet of size 40602 because the schedule clock identifies the phase.

The same trial-cut residuals remain distinct for the complete signed-score function, not merely for an externally emitted event name. Any two of A_0, A_1, A_D have different indicator functions. Take an off-diagonal candidate-validation pair, which belongs to none, and a target-validation pair distinguishing the two events. The resulting scores differ. Thus passing to the validation suffix does not merge the residuals already counted. $\square$

Theorem 29.6 (Matched physical decision complexity and stable certification). *For the two-sphere collision family of Theorem 28.2, let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$, and let $Q_{d,\sigma}$ be its all-on marked target. At the charged training-validation cut of the reset interface,*

$$\kappa_{2/5}(B, Q_{d,\sigma}) = 3. \quad (29.9)$$

More concretely, no auditor retaining at most two states at this cut can have expected score greater than $2/5$ uniformly against all private width-one candidates, at any finite training length, even with feedback-row selection as in Lemma 29.2. One rational auditor with three states at this cut does so with 399 informative training trials and two validation trials. With a redundant ignored training trial, it uses exactly 402 total trials, at most 40602 states at every scheduled epoch, and the exact trial-cut profile (29.8) on its informative training segment.

For fixed positive σ and $d_k \rightarrow d$, the same machine changes its expected score by at most

$$\epsilon_k = \min \left\{ 1, \frac{\sqrt{|d_k - d| + 2700|d_k - d|^2}}{2\sigma} \right\}. \quad (29.10)$$

The private, hidden and visible deficiencies, the no-collision control, and the joint microscopic, likelihood, posterior, innovation and global bounded entropic backward convergence retain all conclusions of Theorems 28.2 and 27.2.

Proof. Theorem 28.2 discharges the physical hypotheses and proves $\|Q_{d,\sigma} - Q_*\|_{\text{TV}} < 1/300$, uniformly on the stated parameters. Theorem 29.1 therefore gives the universal cut-width-two upper bound

$$\mathfrak{V}_2(B, Q_{d,\sigma}) < \frac{3}{8} + \frac{1}{300} < \frac{2}{5}.$$

Lemma 29.2 makes this a bound for the full original feedback-row interface, not only all-on training and not only the constructed rule. The upper construction uses all-on trials and works for every private candidate because only its two all-on parameters enter.

Use the deterministic selector of Lemma 29.4 with $n = 399$, but validate against the actual physical target. For each selected event, changing only the target from Q_* loses less than $1/300$.

Its physical expected score is consequently greater than

$$L_{399} := \sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}. \quad (29.11)$$

Here is a rational verification of the last strict inequality, independent of decimal evaluation. The bounds $e > 2$, $\sqrt{798} > 28$, $\sqrt{2} > 707/500$, and $399/8 > 49$ imply

$$L_{399} > \frac{207}{500} - \frac{1}{112} - \frac{1}{300} - 2^{-51} > \frac{2}{5},$$

since $207/500 - 1/112 - 1/300 - 2/5 = 73/42000 > 2^{-51}$. The number L_{399} is approximately 0.4032891; the rational inequalities, not this approximation, establish the certificate. Its state realization was proved in Theorem 29.5. The general operational classification now yields (29.9) with both inequalities proved for this same physical task.

Changing d leaves the blind candidate source and all its training laws unchanged. The moving-collision estimate and the common-preparation Gaussian theorem bound the target variation by ϵ_k . For any fixed response its target mean changes by at most this amount; averaging over the unchanged training rule proves (29.10). There is no multiplier equal to the number of training trials. The same physical estimates verify the hypotheses of Theorem 27.2; the observation-level density and coupling versions used there are given in Section 26. This transports the same certificate and the bounded conditional-law consumer along the changing microscopic paths. The no-collision regime $d \leq 2/5$ has the vanishing deficiencies already proved in Theorem 28.2; it is not included in the positive-gap parameter interval. $\square$

29.4. What the two exact complexity statements measure. Equation (29.9) minimizes over every auditor at one specified physical decision cut. Equation (29.8) minimizes all trial-cut widths simultaneously for exact implementation of the stated 402-trial rule. These are different optimizations. The first is a task-level optimum, not an artifact of that rule. The second explains the whole online realization of the same rule, not a claim that every successful auditor must have 20301 states somewhere. The sixteen-bit bound includes bit acquisition and validation but does not bound the description length of time-dependent transition tables. It is an implementation upper bound for peak memory. Independent persistent randomization is charged; fresh randomness from the reset experiments is not a hidden cross-cut memory channel.

The former two-count 400-sample construction and its nineteen-bit bound remain in Section 34. They offer a different approximation to the full marked discrepancy. The new three-response rule does not reproduce that rule or attain its oracle envelope. It instead solves the specified physical certification task with an exactly minimal decision-cut alphabet and an explicit full continuation profile. The score is an expectation, not a confidence guarantee from a single realized validation pair.

30. FULL-PROFILE CERTIFICATION OF THE COLLISION FAMILY

The short audit of Theorem 29.6 is retained with its exact continuation profile. We now give a second audit of the same physical class whose entire acquisition-and-validation register is bounded independently of its accuracy parameters. Its cost is a much longer, explicitly counted, preparation schedule.

30.1. The row-indexed decision invariant. For the original row-event interface define

$$\kappa_c^{\text{rows}}(B, Q) = \min\{K : \mathfrak{R}_K^{\text{rows}}(B, Q) > c\}. \quad (30.1)$$

The minimum is $+\infty$ when the set is empty. This is not an implicit change of the one-row symbol κ_c . By Corollary 9.4, it is the exact minimum decision-cut alphabet for a strict score, allowing a finite training schedule and a post-cut distribution on row-event pairs.

Let W_c^{phys} be the minimum peak width among all finite auditors for the original physical row interface, now charging each report-bit, mark, decision and validation epoch. Its clock is externally scheduled, as in the earlier physical implementation. All persistent random selectors are charged.

Theorem 30.1 (A twelve-state physical audit). *In the physical parameter range of Theorem 29.6, put $\beta = 1/300$. For $0 < \theta < 1/4$ and $0 < \delta < 1$, let L, R be at least the respective quantities in (12.1). There is a deterministic audit with $N = 4LR$ candidate training trials and two validation trials whose expected score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2\delta. \quad (30.2)$$

It has at most five states at completed training-trial cuts, ten at intermediate training report-bit cuts, three at the decision cut, and twelve throughout validation. In particular

$$\kappa_{2/5}^{\text{rows}} = 3, \quad 3 \leq W_{2/5}^{\text{phys}} \leq 12. \quad (30.3)$$

One entirely integer schedule takes $\theta = 1/200$, $\delta = 1/1000$, $L = 400$ and $R = 8 \cdot 2^{400}$, so $N = 12800 \cdot 2^{400}$. Its lower bound exceeds $2/5$. The upper bound twelve is not asserted to be the exact minimum integer peak width.

Proof. Use the report-pair probabilities $a = (1-p)(1-q)$ and $c = pq$ from Section 29. In the first $2LR$ training trials run Lemma 12.1 on the indicator of the pair 00. If it returns positive, choose A_1 and ignore the remaining training trials. Otherwise, in the next $2LR$ trials run the same comparison on the indicator of 11, choosing A_0 if positive and A_D if negative. Both phases have fixed lengths, even after an absorbing decision.

A comparison whose parameter is outside its indifference band is correct with probability at least $1 - \delta$. For purposes of analysis, one may define the second comparison on its independent observations even when the implementation ignores them. A union bound gives probability at least $1 - 2\delta$ that neither comparison errs outside its band. Since $\min(a, c) \leq 1/4$ and $\theta < 1/4$, at most one of a, c lies in $[1/2 - \theta, 1/2 + \theta]$.

On this good event the chosen payoff is within θ of

$$E(a, c) = \max\{1 - a - c, 1/2 - a, 1/2 - c\}.$$

Indeed, if one coordinate exceeds $1/2 + \theta$, its associated event is selected and is optimal. If both are below $1/2 - \theta$, A_D is selected and is optimal. In the remaining case the low coordinate's comparison is negative, and choosing either of the two relevant events costs at most the distance of the near coordinate from $1/2$. The range of the three payoffs at any fixed (a, c) is at most one. The expected ideal score is therefore at least $E(a, c) - \theta - 2\delta$. Lemma 29.3 gives $E(a, c) \geq \sqrt{2} - 1$. Changing the target to the physical target loses strictly less than β , proving (30.2).

The first comparator has four states. The second phase has these four states and one extra absorbing label for a previously chosen A_1 , hence five. Retaining the first report bit doubles that count to ten. On the second bit, determine the relevant indicator, update the block state, and discard both bits. The training mark is read and discarded. At the phase boundary the fixed schedule maps the first decision to either the A_1 label or the initial state of the second comparator. At the end it maps the remaining decisions to A_0 or A_D . During validation keep the chosen event, the candidate-event indicator, and a first-bit buffer when needed. This requires at most $3 \cdot 2 \cdot 2 = 12$ states; all events ignore the mark. These are actual phase-compatible transitions, not a bound only on terminal labels.

For the integer schedule, the error estimate in the proof of Lemma 12.1 is $2e^{-8} < 1/1000$. To check it rationally, $e > 8/3$ and $(8/3)^8 > 2000$. Further $\sqrt{2} > 707/500$, and hence

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict score claim without numerical optimization. The training count is exactly $4LR$; early absorption does not delete trials from that count.

For the lower bounds in (30.3), every full-profile auditor has its decision cut among its charged cuts. The feedback pushforward subfamily of Lemma 29.2 bounds any width-two decision continuation by $3/8 + \beta < 2/5$, for every finite adaptive training schedule. The all-on construction just given is a member of the original row interface and has three decision labels. This proves the row-indexed equality and both peak inequalities. $\square$

Corollary 30.2 (Two physical resource regimes). *The same physical score threshold $2/5$ admits the two schedules*

<i>training trials</i>	<i>decision width</i>	<i>peak width</i>
399	3	40602
$12800 \cdot 2^{400}$	3	12

followed by two validations. The first schedule may be padded to 400 training trials as before. For either fixed machine, changing d to d_k at fixed positive noise level changes its expected score by at most ϵ_k in (29.10). If the candidate training kernels also change, their additional cost is the training term in Corollary 12.3.

Proof. The first line is Theorem 29.6; the second is Theorem 30.1. In the moving-collision comparison the candidate training source is identical. Conditional on its chosen event, only one target validation mean changes. The common-preparation Gaussian bound therefore contributes ϵ_k , independently of the length of the unchanged training. The general perturbation formula applies when this equality of sources is absent. $\square$

The long schedule is an existence bound on the resource frontier, not a practical substitute for the 399-trial rule. In an autonomous model the clock must also be stored; Proposition 11.3 gives its explicit additional factor. Both schedules use deterministic transitions, so no precision-dependent randomizer is hidden in the physical upper bound. The one-collision construction remains the verified physical realization. No many-particle or zero-noise theorem is inferred from the larger class of finite experiments.

31. A THIRD RESOURCE REGIME FOR THE SAME PHYSICAL TASK

The three-response obstruction, the short exact selector, and the constant-width block audit concern the same positive-noise marked collision family. A truncated absorbing comparison supplies a further resource pair without changing that family or its threshold.

Theorem 31.1 (Polynomial-horizon physical realization). *Use the original row-event interface and the parameter range of Theorem 30.1. Let $\beta = 1/300$, $0 < \theta < 1/4$, and $m, n \geq 1$. There is a deterministic audit with $2n$ candidate training trials and two validation trials whose expected signed score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2e_{m,n}(\theta). \quad (31.1)$$

It uses at most $2m + 2$ states at completed training-trial cuts, $4m + 4$ at intermediate report-bit cuts, three at the decision cut, and twelve throughout validation. In particular, $\theta = 1/200$, $m = 400$ and $n = 160000$ give score greater than $2/5$, with 320000 training trials and peak width at most 1604.

Proof. Write $a = (1-p)(1-q)$ and $c = pq$ for the actual blind product candidate report probabilities, as in Theorem 30.1. On the first n reports apply Lemma 13.1 to the indicator of 00. If the answer is positive retain A_1 and ignore the second training phase. Otherwise use the next n reports to compare the indicator of 11, retaining A_0 for a positive answer and A_D for a negative answer. Conceptually generate the second phase even when it is ignored. Independence of the

reset preparations gives an error probability at most $2e_{m,n}(\theta)$ for either comparison to be wrong outside its indifference band.

On the complementary event the selected ideal payoff is within θ of $\max(1-a-c, 1/2-a, 1/2-c)$. Indeed $\min(a, c) \leq 1/4$, so at most one coordinate is within θ of $1/2$. If neither is, the two comparisons select the maximizing event; if one is, switching the relevant events changes payoff by at most θ . The three payoffs have range at most one at each (a, c) . Their envelope is at least $\sqrt{2} - 1$ by Lemma 29.3. Replacing the ideal target by the physical target loses strictly less than β and does not change the candidate training law. This proves (31.1).

The walk has $2m + 1$ labels. The second phase adds the previously latched A_1 label, giving $2m + 2$. Retaining the first report bit doubles this number; after the second bit, determine the relevant indicator and discard the pair. The mark is read and discarded. A deterministic clock supplies the phase and the within-phase time. At its end map the remaining walk states to A_0 or A_D . The validation implementation from Theorem 30.1 uses at most twelve states, including the event, first bit, and candidate-event value. No real-valued randomization is needed by this construction.

For the stated integers, $4\theta m = 2n\theta^2 = 8$, so $e_{m,n}(\theta) = 2e^{-8} < 1/1000$. The rational estimates $e > 8/3$, $(8/3)^8 > 2000$, and $\sqrt{2} > 707/500$ give

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict physical margin, with the stated fixed training count and peak width. $\square$

Corollary 31.2 (Three compatible resource bounds). *The original physical task admits all three rows below, each followed by two validations and having decision width three:*

<i>training trials</i>	<i>peak width</i>	<i>training mechanism</i>
399	40602	<i>exact empirical selector</i>
320000	1604	<i>truncated absorbing comparisons</i>
$12800 \cdot 2^{400}$	12	<i>four-state block comparisons.</i>

The all-auditor row-indexed decision optimum remains $\kappa_{2/5}^{\text{rows}} = 3$ and the unrestricted finite-horizon physical peak still satisfies $3 \leq W_{2/5}^{\text{phys}} \leq 12$. The table gives attainable points, not an optimal physical fixed-sample frontier. In particular the 40602 bound is not an all-auditor lower bound for 399 samples.

Proof. Combine Theorems 29.6, 30.1 and 31.1. The feedback-uniform two-state obstruction in Theorem 30.1 continues to apply to every audit in this enlarged set of constructions. $\square$

For a perturbed blind training source with per-active-trial defect α , the two-phase construction also has the bound

$$2 \min\{1, 2\alpha \min(n, m^2)\} + \beta_c + \beta_t$$

on score change, by the proof of Corollary 15.2. The moving-collision parameter alone leaves that source unchanged, so its training contribution is zero; the previous target transport and bounded Gaussian/entropic consumer are retained without change. If one makes the machine autonomous, an explicit bound is $(6n + 12) \max(4m + 4, 12)$ states, since the $2n + 2$ trials each acquire two bits and a mark, with the phase/decision boundaries included. The clock multiplier is an implementation upper bound, not an optimized autonomous-memory claim.

32. CONFIDENCE-LEVEL CERTIFICATION

An expected signed score is an operational separation, not a confidence statement about one validation pair. We now distinguish these two tasks. The null hypothesis below is equality of the candidate and target validation laws; it need not belong to the private alternative family. All

preparations in the theorem are independent resets with one fixed candidate law. The test cannot be given a new adversarial candidate after each reset.

Theorem 32.1 (Confidence lift with a charged accumulator). *Suppose a base reset audit uses n training trials, one candidate and one target validation, and peak width W , with score $D \in \{-1, 0, 1\}$. Assume its expected score is at least $\gamma > 0$ for every alternative. Run J independent copies and reject equality when*

$$\sum_{j=1}^J D_j \geq J\gamma/2. \quad (32.1)$$

Then both the type-I error under equality and the worst-case type-II error are at most $e^{-J\gamma^2/8}$. The whole test has a realization with at most $W(2J+1)$ states and $J(n+2)$ trials. In particular $J \geq 8\gamma^{-2} \log(1/\alpha)$ suffices for both errors to be at most α .

Proof. Under equality, conditional on the training and selected event, the two independent validations have equal means; hence $\mathbb{E}D = 0$. Under each fixed alternative its mean is at least γ . The copies are independent and $D_j \in [-1, 1]$. The exponential-moment estimate for a variable in an interval of length two is $\mathbb{E}e^{s(D_j - \mathbb{E}D_j)} \leq e^{s^2/2}$, by the same tilted-variance argument used in Theorem 9.2. Markov's inequality and minimization in s give $\Pr(J^{-1} \sum (D_j - \mathbb{E}D_j) \geq a) \leq e^{-Ja^2/2}$, and the same lower-tail bound. Set $a = \gamma/2$.

Keep the running integer sum, with at most $2J+1$ values, beside the base machine's register. At each completion add its emitted score and reset only the base register. The sum is not stored in an external history. The clock supplies the repetition index and the fixed terminal threshold; all $J(n+2)$ trials are counted. $\square$

The 399-trial physical audit has mean strictly above $2/5$, so one may use the rational choice $\gamma = 2/5$ in this theorem. This gives the fully explicit sufficient count $J \geq 50 \log(1/\alpha)$ and $401J$ total trials (or $402J$ with the inherited ignored slot). The confidence guarantee uses repetition and its charged sum, not one pair of validation observations.

Theorem 32.2 (One training phase and repeated validation). *For the physical audit of Theorem 30.1, put $\gamma_0 = \sqrt{2} - 1 - 1/300 - \theta > 0$. Retain its chosen event and use m independent candidate-target validation pairs. Reject equality when their score sum is at least $m\gamma_0/2$. The type-I error is at most $e^{-m\gamma_0^2/8}$ and the worst-case type-II error is at most*

$$2\delta + e^{-m\gamma_0^2/8}.$$

The audit uses $4LR + 2m$ trials and at most $12(2m+1)$ states throughout training and validation. Thus $\delta \leq \alpha/4$ and $m \geq 8\gamma_0^{-2} \log(2/\alpha)$ suffice for error at most α .

Proof. The proof of Theorem 30.1 gives an event of training probability at least $1 - 2\delta$ on which the conditional mean score of the selected event is strictly greater than γ_0 . Conditional on that event and its selected label, the validation pairs are independent. Apply the tail bound in Theorem 32.1 conditionally and add the training failure probability. Under equality every selected event has zero mean, regardless of training, so no training failure term is needed for type-I error. Keep the three-valued event label, the within-pair indicator and first-bit buffer, and a sum with $2m+1$ values. Training itself needs no more than ten states, so the displayed bound covers every phase. $\square$

The tests in this section have a different output task from maximizing an expected witness score. The exact three-state decision-cut theorem is not asserted to be an optimal memory theorem for these binary tests. For perturbations of training and validation laws, the coupling argument of Theorem 11.2 bounds changes of each rejection probability by the sum of the per-call total variation errors. In a source-preserving physical transport, alternatively lower the conditional

mean margin by the single-row target variation and reapply the conditional tail estimate. This latter argument transports the confidence guarantee without pretending that an entire m -sample validation law has the same total variation distance as one validation observation.

33. CONFIDENCE TESTS WITH PRICED FINITE-STATE AMPLIFICATION

The signed-sum construction in Section 32 converts expectation into confidence. There is also a direct sample–memory choice for the amplifier itself. Its formulation keeps the binary testing objective distinct from the expected-score task.

Theorem 33.1 (Two finite-state confidence amplifiers). *Suppose a reset audit uses b training trials, two validations, peak width $W \geq 3$, and reports $D \in \{-1, 0, 1\}$. Under the equality null $\mathbb{E}D = 0$, and under every alternative in a fixed class $\mathbb{E}D \geq \gamma > 0$. Fix a known dyadic $0 < g \leq \gamma$, and put $\theta = g/8$.*

- (i) *For integers $m, n \geq 1$, there is a binary test with both errors at most $e_{m,n}(\theta)$, using $n(b+2)$ preparations and peak width at most $(2m+1)\max(W, 9)$. Thus both errors are at most α with the choices (13.5), with $\delta = \alpha$.*
- (ii) *With L, R as in (12.1), using $\delta = \alpha$ and the same θ , there is another test with both errors at most α , $2LR(b+2)$ preparations, and peak width at most $4\max(W, 9)$. Its very long external schedule is charged separately. The confidence error, not the input parameter, determines these read-only horizon constants.*

Proof. Run independent copies of the base audit with fresh preparations and fresh private bits. Conditional on each output D , generate an additional coin Z with probability

$$\mathbb{P}(Z = 1 \mid D) = \frac{1}{2} + \frac{D - g/2}{4}. \quad (33.1)$$

This is between zero and one, since $0 < g \leq \gamma \leq 1$ and $-1 \leq D \leq 1$. If g has denominator 2^s , the probability has denominator dividing 2^{s+3} and is sampled with $s+3$ fair bits. The coins are independent under each fixed parameter. Their mean is $1/2 - g/8$ under the null and at least $1/2 + g/8$ under the alternatives. Apply the walk of Lemma 13.1, rejecting the null for a positive result, to prove (i). Apply the block comparison of Lemma 12.1 to prove (ii).

At every base-audit epoch retain the amplifier state alongside its W labels. After obtaining D , retain that value and at most three Bernoulli-sampler labels while drawing (33.1). This requires at most nine labels per amplifier state. Update the amplifier and reset the base audit before the next repetition. The terminal rejection decision is a function of the amplifier state and requires no additional persistent random seed. This proves the stated full-cut peak bounds. Even after the amplifier is absorbed, padded base-audit repetitions remain charged to the preparation count. The clock supplying the repetition, base-audit epoch and fair-bit depth is external; storing it multiplies the displayed bound by at most one plus the total number of such epochs. $\square$

For the physical expected-score theorem one may take $g = 3/8$, since $3/8 < 2/5$. The additional coin then needs at most six fair bits. Thus confidence can be purchased either with a growing absorbing counter and polynomial repetition count, or with a constant register and a much longer clocked schedule. Neither statement identifies the minimum state count of the binary testing problem with the three conditional validation responses needed by the expected-score problem. The signed-accumulator and train-once confidence theorems remain available as different resource regimes.

The output-law transfer in Theorem 15.1 applies to a completed binary test. In particular, if perturbations change its output law by at most ε , its type-I and type-II bounds each increase by at most ε . Transporting all repeated raw data instead counts every affected preparation, as in Section 32. Independence of the base repetitions and uniformity over the fixed alternative class

are assumptions of the amplification theorem; adaptation of the alternative between repetitions is not being asserted.

34. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 18.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 28, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 34.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,*

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{TV} - n^{-1/2}. \quad (34.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250), \min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (34.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass $5/16$ when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is

known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 8.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 34.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 28.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor's register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 28.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 27.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\varepsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 28.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 34.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 23.1 gives

$$\min_{p,q} \|Q_* - b(p, q)\|_{\text{TV}} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 28.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (34.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 34.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

For changing d_k , Lemma 28.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 25.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 18.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 27.2, proving the global backward assertions. Finally the audit uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure

theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

35. EXPLICIT PHASE TRANSITIONS FOR THE TWO-COUNT CERTIFICATE

For comparison with the new residual quotient, we give the complete phase-state specification of Lemma 34.1. It realizes the original 400-training-trial rule exactly, not the majority rule of Section 29. Let

$$C = (C_0, C_1) \in \{0, \dots, 250\}^2, \quad \tau(x) = \min(x, 250).$$

The free clock records the training index, the within-trial observation epoch, the validation phase, and deterministic processing epochs. It contains no data-dependent information. At the start, $C = (0, 0)$.

For a completed transcript (x_1, x_2, w) define the deterministic update

$$T(C, x_1, x_2, w) = (\tau(C_0 + \mathbf{1}_{\{(x_1, x_2) = (0, 0)\}}), \tau(C_1 + \mathbf{1}_{\{(x_1, x_2) = (1, 1)\}})).$$

The value of w does not enter this training update. The phase transitions are the following; x denotes the current input, not an earlier unretained report.

Phase	Retained data	Next transition
Training start	C	Read x_1 and keep (C, x_1) .
Second report	(C, x_1)	Read x_2 and keep (C, x_1, x_2) .
Terminal mark	(C, x_1, x_2)	Read w and keep (C, x_1, x_2, w) .
Count update	(C, x_1, x_2, w)	Replace by $T(C, x_1, x_2, w)$; discard the transcript.
Event formation	C	Replace by the event mask $A(C)$ defined below.
Candidate validation	A , partial transcript	Complete the transcript; replace it by $i = \mathbf{1}_A(x_1, x_2, w)$.
Target validation	(A, i) , partial transcript	Complete the transcript and emit $\mathbf{1}_A(y_1, y_2, v) - i$.
Score	A value in $\{-1, 0, 1\}$	Terminate.

The event formation map, including both threshold ties, is

$$\begin{aligned} 000 \in A(C) &\iff C_0 < 250, & 001 \in A(C) &\iff C_0 < 150, \\ 110 \in A(C) &\iff C_1 < 150, & 111 \in A(C) &\iff C_1 < 250. \end{aligned}$$

No other atom is included. For each report pair there are three choices: both marked atoms, only its larger-target-mass atom, or neither. Thus there are at most nine masks. Clipping at 250

preserves every comparison, including equality; no flag for a count exceeding the clipping threshold is required.

At a training epoch the possible states are bounded by $8 \cdot 251^2 = 504008$: the factor eight dominates every partial or complete transcript buffer. At validation epochs the largest alphabet has at most $9 \cdot 2 \cdot 8 = 144$ values, including the already computed candidate-event indicator. Event formation and final emission need at most nine and three states, respectively. Embed each phase alphabet into $[504008]$ by a fixed injective encoding. Because the phase is supplied by the free clock, these encodings need not have disjoint images. The next row decodes the current phase’s encoding and applies the displayed transition. This gives formal deterministic rows with entries zero or one at every epoch. The buffers, event mask, and validation indicator are all charged. No unpriced data store is introduced by label reuse.

APPENDIX A. THE PROOF DEPENDENCY CHAIN

An arrow below means an actual use of a theorem in the displayed proof, not an assertion of priority or an artificial dependency between papers. The preserved exact physical lower and upper bounds meet at the decision cut. The new full-profile theorem supplies an explicit training realization of general decision covers; its proof does not invoke the historical aggregate kinetic claims.

Input	Consequence using that input
Marked private rows and normalized continuations, Theorems 7.2, 8.2	The product behavior image and its validation continuation responses.
Decision continuation spectrum, Theorem 9.2	Universal two-state physical obstruction, Theorem 29.1; the strict-cut classification (29.9).
Product constraint (29.1) and three-response envelope, Lemma 29.3	Uniform finite-sample score, Lemma 29.4.
Finite-valued residual realization, Theorem 16.1	Exact full training profile and bit-level construction, Theorem 29.5.
Marked scattering and moving-collision estimate, Theorem 28.2	Both physical cut bounds and transport of the very same auditor, Theorem 29.6.
Common-preparation Gaussian comparison, Theorem 25.1; the three version lemmas in Section 26	The global conditional-law consumer, Theorem 27.2, on that moving-collision family.

The prior binary audit theorem remains a separate calibration, not a premise of the physical memory lower bound. The original two-count certificate remains a separate full-discrepancy approximation, not a claimed optimal implementation. The source-preserving build, rational checks and finite enumerations accompanying this article test the stated implementation and identities; they are not independent analytic proof certification.

A.1. The full-profile resource chain. The new central arrow is

strict response cover $\longrightarrow$ finite block tournament $\longrightarrow$ rational whole-schedule auditor.

Theorem 12.2 uses the decision spectrum, the finite block lemma and the dyadic sampler. It does not use a training oracle. Theorems 11.1 and 11.2 track approximation, calibration and precision on complete causal schedules. Theorem 17.1 uses one shared capacity budget for all cuts, and Theorem 32.1 retains the score accumulator. Together these establish a finite controlled-experiment instance of the program’s resource–precision–statistical-error objective. A universal singular-modulus rate or an infinite-horizon kinetic realization is not an input or a consequence of this finite theorem.

The current C2-scoped consumer remains the bounded, common-preparation, positive-noise Gaussian/entropic theorem. The original B4 normalized nonlinear resolvent formula cannot be inherited: its displayed linear difference identity fails on constant inputs. For a separately justified graph H with single-valued $R_\lambda = (I - H/\lambda)^{-1}$, the correctly typed identity is instead

$$R_\lambda h = R_\mu [(\lambda/\mu)h + (1 - \lambda/\mu)R_\lambda h].$$

Indeed both sides solve the same μ -resolvent inclusion. This conditional identity does not establish the graph's range or its kinetic corrector. Those historical targets remain in the program rather than being assigned proof credit from a finite resource theorem.

APPENDIX B. A NAMESPACE MAP OF THE ELEVEN-COMPONENT PROGRAM

The namespaces in this map separate historical paper targets from GTF statements proved here. Every historical component is retained. An interface means that the indicated consumer can be applied after its model-specific hypotheses are proved; it is not an automatic proof of the historical component’s aggregate theorem.

Paper	Retained historical role	GTF relation and precise proof credit
A1	Protocol and finite task reduction	Marked comparison, priced composition and decision-cut certification; no automatic all-order physical response conclusion.
A2	Independent primary geometric chain	Its protocol adapter is historical; the primary geometric chain is not a premise or consequence of a theorem in this article.
A3	Dynamical/potential-space targets	Conditional comparison interface; no new assertion of the historical dynamical closure.
A4	Conditional kernels and forgetting	Historical finite-model consumer; common-preparation posterior estimates feed the current C2-scoped consumer when their hypotheses hold.
B1	Microscopic realization targets	Declared marked experiment interface; the present scattering construction is fixed-particle.
B2	Dynamic action and kinetic targets	Historical action estimates are not used to prove the present decision-cut theorem.
B3	Balance, response and corrector targets	No current proof depends on the historical aggregate balance/kinetic claims.
B4	Nonlinear kinetic semigroup	The historical normalized-resolvent identity and its aggregate downstream claims are not premises here. Their independent repair remains a program obligation.
C1	Statistical acquisition/inference	Original behavior image, finite-reset experiments, and the present response-envelope and microscopic certification consumers.
C2	Duals, response, memory and projections	The current bounded Gaussian/entropic transport replaces the optional-projection step only in its explicitly proved class; it does not certify the historical strict-dual, form-response or rigidity aggregate.
D1	Phase and compact statistical targets	Historical finite-model consumer retained; no new complete phase-theory closure is inferred.

The machine-readable companion records the exact predecessor graph and source identities, including the original B4 and C2 statements inspected at their frozen commit. Names such as `gtf20:decision-cut` and `historical:B4:r17-main` denote different proof obligations. The organizing theorem is a general comparison-and-certification result with an essential physical consumer. It does not replace the entire historical pipeline by a declaration that every aggregate theorem follows from it. The broader program remains intact as a mathematical program.

APPENDIX C. PRESERVED ORGANIZING STATEMENTS

The following two statements retain the earlier organizing labels and claims for cross-reference. Their full proofs are included in the canonical modules above. The unchanged predecessor article and its complete historical development are appended in the separate companion, so that reorganization does not discard earlier derivations.

Theorem C.1 (Decision continuation complexity and physical certification). *The following assertions hold with all persistent data charged to the appropriate machine.*

(i) *The supremum $v_{n,K}$ of worst-case expected scores with n training samples and K states at the decision cut satisfies*

$$0 \leq \mathfrak{V}_K - v_{n,K} \leq \min \left\{ \sqrt{(|\Omega| - 1)/n}, \sqrt{2 \log(2K)/n} \right\}.$$

The least K permitting a strict score greater than c at some finite training length is the least size of a cover of B by sets $\{b : (Q - b)h > c\}$. This spectrum is Lipschitz under uniform total variation perturbations of candidate and target laws.

(ii) *For the ideal marked private product family B of the two-sphere experiment, $\mathfrak{V}_1 = 1/8$ and $\mathfrak{V}_2 = 3/8$. Throughout the collision interval $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the minimum decision-cut width for uniform physical score greater than $2/5$ is exactly three. The lower bound applies to every finite-reset auditor with the specified interface, not merely a particular counter implementation; feedback-row selection does not evade it.*

(iii) *One deterministic rational rule has physical score greater than*

$$\sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}.$$

It uses 399 informative training trials, an optional ignored training trial and two independent validations, hence fits the 402-trial schedule. It has exactly three decision-cut states. On its informative training segment, the minimum simultaneous widths for exact stochastic implementation are

$$r_t = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq 199, \\ \binom{402-t}{2}, & 200 \leq t \leq 399. \end{cases}$$

Their maximum is 20301. Including individual report-bit acquisition and the validations, at most $40602 < 2^{16}$ retained states suffice.

(iv) *At an arbitrary cut of a finite-valued decision, a weighted graph of distinguishing suffixes gives a distribution-sensitive error bound through its minimum monochromatic weight under K labels. Applied to Bayes margins, it yields a quantitative constrained-audit regret bound. At zero error it recovers the exact simultaneous residual profile, even for stochastic implementations.*

(v) *For fixed positive σ and changing d , the same physical rule has the target-variation modulus $\min\{1, \sqrt{|d' - d| + 2700|d' - d|^2/(2\sigma)}\}$, with no training-count factor. The microscopic paths, likelihoods, posterior processes, innovations, and bounded entropic backward values, integrands and martingale integrals have the joint transport conclusions proved below. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. In the no-collision regime $d \leq 2/5$ all three finite-experiment deficiencies vanish.*

Proof. Part (i) is Theorem 9.2. The exact ideal spectral values are Theorem 29.1; Lemmas 29.3 and 29.4 and Theorem 29.6 prove the matching physical construction. Theorem 29.5 proves (iii), including all acquisition and validation phases. Theorem 16.1 and Corollary 16.2 prove (iv). The microscopic estimates in Theorem 28.2, the explicit coupling-version lemmas in Section 26, and Theorem 27.2 prove (v). $\square$

Theorem C.2 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

(i) *A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing*

machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+ / (r - 1)$.

(ii) For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (19.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

(iii) In the prepared two-sphere experiment of Theorem 34.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.

(iv) With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .

C.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem C.2; the earlier 2800-trial audit remains valid.

Theorem C.3 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K, N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It

also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) In the two-sphere model of Theorem 28.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.

APPENDIX D. UNRESTRICTED-HORIZON MULTICUT REALIZATION

The following organizing statement retains the unrestricted finite-horizon comparison and its complete proof dependencies. The interfaces are those fixed in the introduction.

Theorem D.1 (Multicut realization and certification). *The following statements hold under the interfaces and resource conventions just described.*

(i) For $d = |\Omega|$, let W_c be the minimum peak width over all finite training schedules, including every random-bit and validation cut, and let W_c^{rat} restrict to finite fair-bit implementations. Then

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q).$$

For a K -response cover, a finite-horizon tournament realizes this bound with $2LR(K-1)$ training observations, where $L = \lceil \log(2/\delta)/(4\theta) \rceil$ and $R = \lceil 2^L \log(2/\delta) \rceil$. Its uniform score loss is at most $2^{1-r} + 2\rho + 4(K-1)\theta + 2(K-1)\delta$, with dyadic precision r and response-mean calibration error ρ .

(ii) Approximate normalized continuation lists with compatible one-step updates and local causal errors η_t realize their entire width profile with total error at most $\sum_t \eta_t$. Full-profile risk bounds also charge the sum of the per-call calibration errors and the row-rounding errors. Composition multiplies the simultaneous simulator and auditor profiles. A shared word-capacity constraint turns weighted residual obstructions at several cuts into a single Bayesian regret lower bound. A family of point-source experiments has exact decision spectra and matching linear peak-width growth; overlapping independent retention tasks have simultaneous sharp exact profiles and quantitative approximate lower bounds.

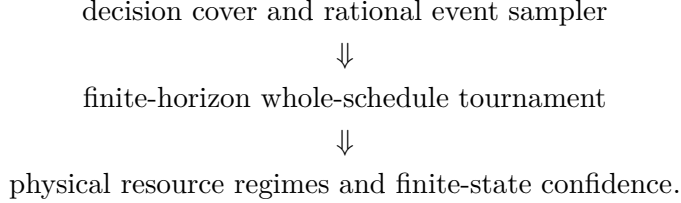
(iii) In the prepared two-sphere collision family, for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the row-indexed decision complexity for score greater than $2/5$ is exactly three. A deterministic physical audit has peak width at most twelve throughout report-bit acquisition and validation, with an explicit finite preparation count. The earlier 399-training-trial audit is also retained, with peak width 40602 and its exact simultaneous residual profile. For the complete physical task these give $3 \leq W_{2/5}^{\text{phys}} \leq 12$; the upper endpoint is not claimed to be an exact integer optimum.

(iv) Independent repetition turns either positive expected-score certificate into a test with both errors at most α , charging the score accumulator and every preparation. The short physical audit needs $J \geq 50 \log(1/\alpha)$ repetitions. The physical certificate and the bounded common-preparation Gaussian/entropic conditional-law consumer are stable under the moving-collision perturbations specified below.

Proof. Part (i) is Theorem 12.2. Part (ii) follows from Theorems 11.1, 11.2, 17.1, 17.2, 17.3 and Proposition 11.3. Part (iii) is Theorem 30.1 together with the preserved Theorems 29.5 and 29.6. Part (iv) follows from Theorems 32.1 and 32.2, Corollary 30.2, and the common-preparation transport results in Theorems 28.2 and 27.2. $\square$

APPENDIX E. DEPENDENCIES OF THE STATISTICAL RESOURCE FRONTIER

The general preparation bound uses the decision spectrum only to supply a finite response list and its strict margin. Dyadic event realization supplies its executable randomizers. The absorbing-walk lemma then supplies both a finite comparison error and the expected number of active updates. Thus



The occupation-weighted transfer theorem acts on the induced retained kernels of each implemented machine. In the independent reset setting it consumes the walk's active-time estimate. Its proof does not invoke any claim about convergence of changing filtrations.

The exact revelation theorem is an independent, fully solved finite statistical model inside the same profile-constrained decision layer. Its necessary direction applies to every stochastic machine; its sufficient direction constructs compatible nested labels. It is not used as a physical collision lower bound. The lower bound three in that family still comes from the original feedback-row obstruction. Likewise, the complete finite-word residual profile of the short physical selector is retained as an exact implementation theorem and not relabeled as a statistical optimum.

These distinctions give an actual resource–precision–error consumer for the foundation blueprint: preparation and state budgets come from Theorem 13.2, calibration and rationalization from its explicit coin construction, perturbation from Theorem 15.1, and a confidence-level readout from Theorem 33.1. The exact frontier provides a sharp case of the same optimization with nonconstant alphabet and arbitrary state profiles. None of these arrows presupposes the historical B4 kinetic range theorem, the broader C2 rigidity aggregate, or an unproved dependency of the independent A2 chain.

APPENDIX F. THE CONSTRUCTIVE RESOURCE THEOREM

The constructive statement and its proof are retained here as a single reference point for the supporting development.

Theorem F.1 (Continuation profiles and statistical resources). *The following statements hold in the declared clocked finite-state model.*

(i) *Every strict K -response cover of envelope v has, for integers $m, n, r \geq 1$ and dyadic calibration error ρ , a rational whole-schedule realization with*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1),$$

and worst-case score at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)(e^{-4\theta m} + e^{-2n\theta^2}).$$

Here $K \geq 2$; one response needs no training. The previously available constant-width realization gives, in the same model, $\kappa_c \leq W_c \leq W_c^{\text{rat}} \leq 12d\kappa_c$.

(ii) *Suppose a parameter in $\{1, \dots, M\}$ has known ordered prior π , and report t independently reveals that parameter with probability q_t and otherwise reports $*$. Among all stochastic machines having at most K_t labels after each report, the exact Bayesian success probability is*

$$a\pi_1 + \sum_{t=1}^n w_t \sum_{j=1}^{k_t} \pi_j, \quad k_t = \min(M, K_t, \dots, K_n),$$

where $a = \prod_t(1 - q_t)$ and $w_t = q_t \prod_{s>t}(1 - q_s)$. One compatible deterministic machine attains all the cut bounds simultaneously. The formula yields exact sample–peak tradeoffs and exact losses under checkpoint insertion.

(iii) If P_t, Q_t are the induced retained-state transition kernels of a fixed execution with common initial law, its terminal-law change is bounded by

$$\min \left(1, \sum_t \mathbb{E}_P \|P_t(X_{t-1}, \cdot) - Q_t(X_{t-1}, \cdot)\|_{\text{TV}} \right).$$

For absorbed reset comparisons this prices active updates even when training laws differ. It does not remove ignored preparations from the schedule or from the full raw-data law.

(iv) In the marked positive-noise collision experiment, a 320000-training-trial audit with peak width at most 1604 has score greater than $2/5$. The 399-trial, 40602-state audit and the 12-state long-clock audit are also retained. The exact row-indexed decision width remains three. Separate finite-state amplifiers yield specified type-I and type-II errors, with a polynomial-horizon growing counter or a constant register and a longer clocked schedule.

Proof. Part (i) is Theorem 13.2 together with Theorem 12.2. Part (ii) is Theorem 14.1 and its corollaries. Part (iii) is Theorem 15.1 and Corollary 15.2. Part (iv) follows from Theorem 31.1, Corollary 31.2, and Theorem 33.1. $\square$

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